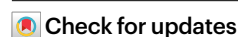


The BRUTUS iron sensor and E3 ligase facilitates soybean root nodulation by monoubiquitination of NSP1

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Legumes form root nodules with symbiotic nitrogen-fixing rhizobacteria, which require ample iron to ensure symbiosis establishment and efficient nitrogen fixation. The functions and mechanisms of iron in nitrogen-fixing nodules are well established. However, the role of iron and the mechanisms by which legumes sense iron and incorporate this cue into nodulation signalling pathways remain unclear. Here we show that iron is a key driver of nodulation because symbiotic nodules cannot form without iron, even under conditions of sufficient light and low nitrogen. We further identify an iron optimum for soybean nodulation and the iron sensor BRUTUS A (BTSa) which acts as a hub for integrating iron and nodulation cues. BTSa is induced by rhizobia, binds to and is stabilized by iron. In turn, BTSa stabilizes and enhances the transcriptional activation activity of pro-nodulation transcription factor NSP1a by monoubiquitination from its RING domain and consequently activates nodulation signalling. Monoubiquitination of NSP1 by BTS is conserved in legumes to trigger nodulation under iron sufficiency. Thus, iron status is an essential cue to trigger nodulation and BTSa integrates cues from rhizobial infection and iron status to orchestrate host responses towards establishing symbiotic nitrogen fixation.

Symbiotic nitrogen fixation between legume plants and nitrogen-fixing rhizobacteria is the largest potential natural source of new nitrogen in the biosphere, with implications for global primary production, carbon sequestration and element cycling^{1–4}. Under low-nitrogen conditions, legumes release flavonoid compounds to induce rhizobia to produce and secrete lipochitooligosaccharides (LCOs), known as nodulation (Nod) factors. Nod factors are sensed on the plasma membrane of root-hair cells by host Nod-factor receptors^{5–10}. The perception of Nod factors by the LysM receptor kinase complexes triggers the nodulation signalling cascade by inducing nuclear calcium spiking, decoding of the calcium signals and activating a series of transcription factors, including NODULATION SIGNALLING PATHWAY 1/2 (NSP1/2) and its

downstream nodulation-essential gene *NODULE INCEPTION (NIN)*, which controls all known nodulation processes^{11–19}. Rhizobia then infect and colonize root cells to form root nodules wherein rhizobial nitrogenases reduce dinitrogen to ammonia to provide the host plant with sufficient nitrogen, in exchange for organic molecules from photosynthesis^{20,21}. The formation and maintenance of nodule functions require substantial resources, and host plants tightly control nodule numbers through a root–shoot long-distance negative regulatory system called autoregulation of nodulation (AON)^{22–29}. However, nodule formation and function are also influenced by environmental factors, such as light and nutrients, which provide carbon and essential elements for symbiotic nitrogen fixation^{1,30–37}.

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Iron (Fe) is a vital micronutrient for all forms of life. Despite being the second-most-abundant metal element on our planet, iron mainly exists in the highly insoluble ferric form in most habitats, making it difficult for bacteria and plants to absorb and use directly³⁸. Iron is a key component or cofactor of various proteins or enzymes in plant cells and plays pivotal roles in diverse processes, such as respiration, photosynthesis and metabolism of reactive oxygen and nitrogen species³⁹. In legumes, both nitrogenase in nitrogen-fixing bacteria and leghaemoglobin in infected nodule cells are iron-containing proteins; thus iron levels directly regulate nitrogenase activity, thereby influencing the efficiency of symbiotic nitrogen fixation^{40,41}. While the link between symbiotic nitrogen fixation in root nodules and iron has been well documented in legumes, the roles and mechanisms of iron during the early stages of nodulation are still not fully understood. Limited research has suggested that certain legumes, such as *Lupinus angustifolius*, do not form nodules in iron-deficient environments, highlighting the essential role of iron in rhizobia–host interaction, infection and nodule development^{42–44}. Inadequate iron levels have been shown to reduce nodule formation in different legume species, such as soybean (*Glycine max*)^{36,45}. The effects of varying concentrations of iron on nodulation and the optimal iron requirement for nodulation have not been comprehensively studied yet.

To cope with fluctuations in environmental iron levels, plants have evolved mechanisms to sense iron status and modulate the expression of genes involved in iron absorption and transport. This is achieved via a transcriptional regulatory network consisting of a set of transcription factors (TFs)⁴⁶. In *Arabidopsis*, under iron deficiency, FER-LIKE IRON-DEFICIENCY-INDUCED TRANSCRIPTION FACTOR 1 (FIT1) interacts with Ib-subgroup bHLH TFs (bHLH38/39/100/101) to regulate iron uptake-related genes such as *FERRIC REDUCTASE-OXIDASE 2* (*FRO2*) and *IRON-REGULATED TRANSPORTER 1* (*IRT1*)^{47–50}. The expression of *FIT* and Ib-bHLH TFs is induced by upstream IVb-subgroup bHLH TFs (bHLH121/URI) and IVc-subgroup bHLH TFs (for example, bHLH105/ILR3 and bHLH115)^{51–54}. bHLH121/URI is phosphorylated and interacts with subgroup-IVc bHLH TFs to induce Ib-bHLH TFs, *POPEYE* (*PYE*), *BRUTUS* (*BTS*) and *BRUTUS-LIKE* (*BTSL*), to activate the iron-deficiency response^{53,54}. *BTS*/*BTSL* are E3 ligases that contain Fe-binding haemerythrin (HHE) domains in the N terminus, a CHY zinc-finger domain and a RING domain in the C terminus^{55,56}. When sufficient iron is available, *BTS* and *BTSL* bind to iron and interact with and ubiquitinate target proteins, including IVc bHLH TFs, bHLH121/URI and *FIT*, leading to their proteasomal degradation, thereby turning off the Fe-deficiency response^{53,54,56–58}. In addition, iron-deficiency-induced IRON MAN (IMA) peptides compete with bHLH105/bHLH115 to interact with *BTS*, preventing the degradation of these TFs by *BTS* and thereby triggering the response to iron deficiency⁵⁹. Thus, *BTS* proteins function as potential iron sensors, playing an indispensable role in maintaining iron homeostasis during plant growth and development⁶⁰.

Iron plays a central role in the interaction between plants and microbes, as both require iron^{61,62}. For symbiotic nodulation in *Lotus japonicus*, IMA is activated by NIN to regulate iron provision⁶³, and CHY1/2 and LJRING, the homologues of At*BTS* in soybean and *L. japonicus*, function in nodulation^{64,65}, suggesting a link between iron and

nodulation signalling. However, it is unclear exactly how the host senses iron levels to orchestrate effective rhizobia–host interactions and trigger symbiotic nodulation. Here we present an in-depth analysis of nodulation in soybean in response to iron availability. We sought to address the following questions: what is the role of iron in symbiotic nodulation; what determines sensing of iron availability during nodulation; how does the iron signal integrate to the nodulation signalling pathway; and whether the mechanism of legume nodulation in response to iron is conserved. Our goal is to determine the role of iron in nodule initiation and uncover the molecular mechanism exploited by soybean to monitor external iron levels, which guarantees the formation of nodules under favourable iron conditions.

Results

Iron is essential for soybean root nodulation

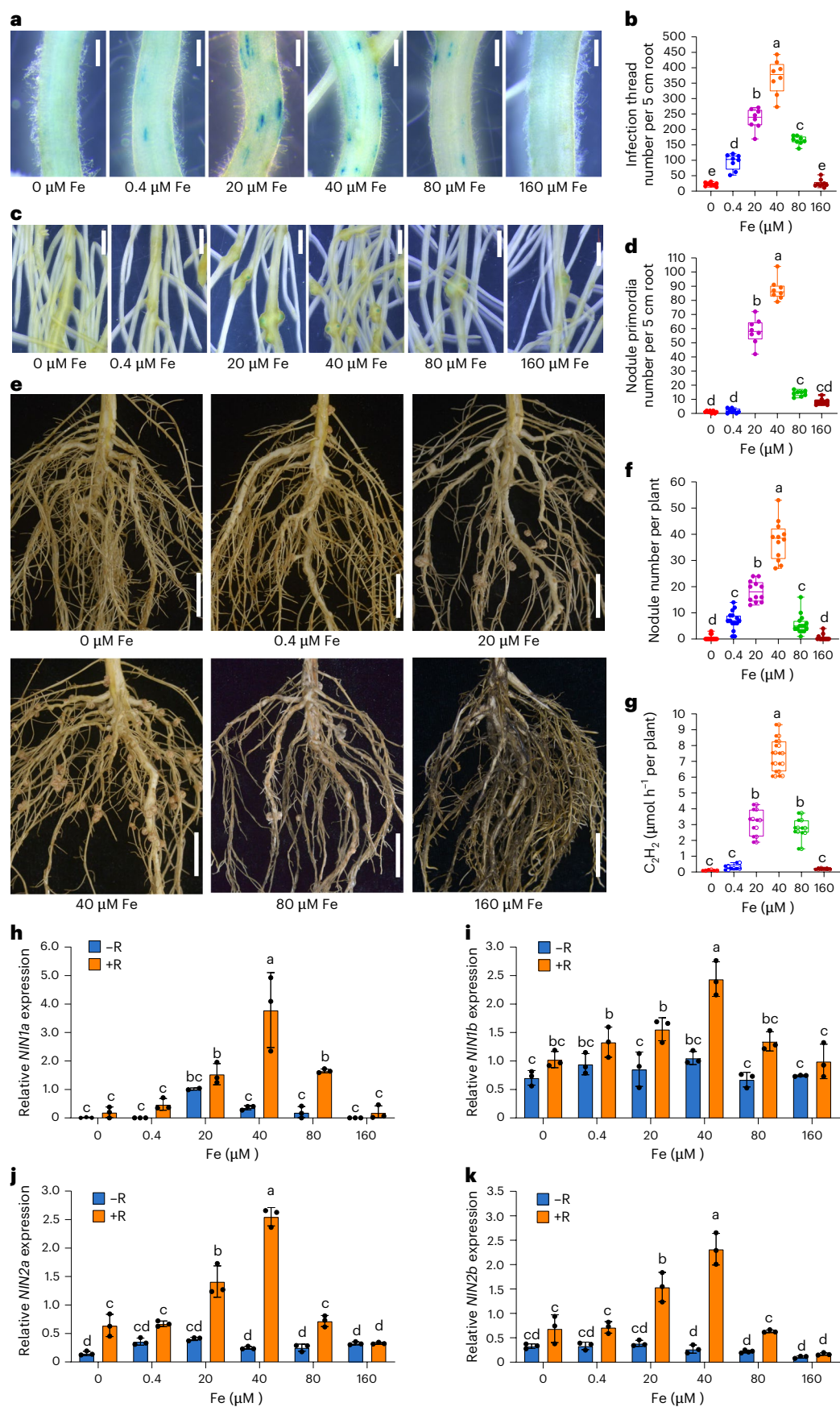
To determine the role of iron in soybean nodulation, we initially investigated its effect on nodulation. We inoculated the soybean reference cultivar Williams 82 (Wm82) with *Bradyrhizobium diazoefficiens* USDA110 rhizobia in a hydroponic growth setup with iron (Fe–EDTA) concentrations of 0, 0.4, 40 and 400 μM under low-nitrogen and sufficient light conditions. Plants formed the highest number of nodules at 40 μM iron, and almost no nodules at 0 and 400 μM iron. Consistently, infection number and primordia number were dramatically decreased under 0 and 400 μM iron (Extended Data Fig. 1a–f). To ascertain the optimal iron concentration for nodulation, we then conducted a comprehensive assessment of iron concentration, encompassing 0, 0.4, 20, 40, 80 and 160 μM , to evaluate their effects on nodulation (Extended Data Fig. 1g–n). The number of infection threads, nodule primordia, nodules and nitrogenase activity all increased up to and peaked at 40 μM but dramatically decreased at 80 μM Fe (Fig. 1a–g). Nodules barely formed at 160 μM Fe, although a few infection threads and primordia were observed (Fig. 1a–g). Our results indicate that iron is essential for soybean nodulation, with 40 μM being an optimal iron concentration, at least under this setup. Excess iron led to accumulation of reactive oxygen species (ROS) in roots (Extended Data Fig. 1h–j), potentially causing molecular damage and cell toxicity that disrupt nodulation. This aligns with previous findings on increased ROS and cytotoxicity induced by excess iron via the Fenton reaction in plant cells⁶⁶.

Expression of *NIN* genes mirrored nodulation phenotypes (Fig. 1h–k). Genes upstream of *NIN1a*, namely, *NSP1a/b* and *NSP2a/b*, the functional orthologues of central nodulation regulators *NSP1* and *NSP2* in other legumes, were minimally influenced by these iron concentrations (Extended Data Fig. 1k–n). Together, these data suggest that iron is a strong driver of the soybean–rhizobia symbiosis and indicate that soybean can sense iron status and transmit this to regulate the core nodulation gene *NIN1a*, thereby controlling nodulation. Next, we aimed to investigate the response of soybean nodulation to low iron. Although a hydroponic approach enables precise control over iron concentrations, due to its complexity in operation and its negative impact on plant growth, we eventually chose to carry out follow-up experiments using a 200 μM AA-DP/2,2'-bipyridine-simulated low-iron method (~20 μM iron) in soil, which yielded a similar effect on soybean nodulation³⁶.

Fig. 1 | Iron levels determine root nodulation in soybean. **a**, Representative images of infection threads in hydroponically grown wild-type Wm82 roots with GUS-tagged USDA110 rhizobia under different iron concentrations at 3 DAI. Scale bars, 200 μm . **b**, Number of infection threads under the indicated iron treatments. Data are means $\pm$ s.d.; $n = 8$ (for 0, 0.4, 20, 40, 80 μM Fe) and 9 (for 160 μM Fe). **c**, Representative images of nodule primordia in Wm82 roots with GUS-tagged USDA110 rhizobia under the indicated iron conditions at 6 DAI. Scale bars, 1 mm. **d**, Number of nodule primordia under the indicated iron treatments. Data are means $\pm$ s.d.; $n = 8$. **e**, Representative images of Wm82 nodules with GUS-tagged USDA110 rhizobia under the indicated iron conditions at 21 DAI. Scale bars, 5 mm. **f**, Nodule numbers under the indicated iron treatments. Data

are means $\pm$ s.d.; $n = 15, 15, 12, 12, 15, 12$ (for 0, 0.4, 20, 40, 80, 160 μM Fe).

g, Nitrogenase activity under the indicated iron treatments. Data are means $\pm$ s.d.; $n = 3, 5, 7, 11, 7, 5$ (for 0, 0.4, 20, 40, 80, 160 μM Fe). **h–k**, RT-qPCR of *NIN1a* (**h**), *NIN1b* (**i**), *NIN2a* (**j**) and *NIN2b* (**k**) expression under the indicated iron conditions at 1 DAI. Data are means $\pm$ s.d.; $n = 3$ and relative expression was normalized to *ELF1b*. For boxplots (**b, d, f, g**), whiskers indicate minimum and maximum values, centre lines indicate medians, and box boundaries indicate the upper (25th percentile) and lower (75th percentile) quartiles. Different lowercase letters above plots indicate significant differences determined using one-way ANOVA ($P < 0.05$) (**b, d, f, g**) or two-way ANOVA ($P < 0.05$) (**h–k**). The experiments were repeated 3 times with similar results.



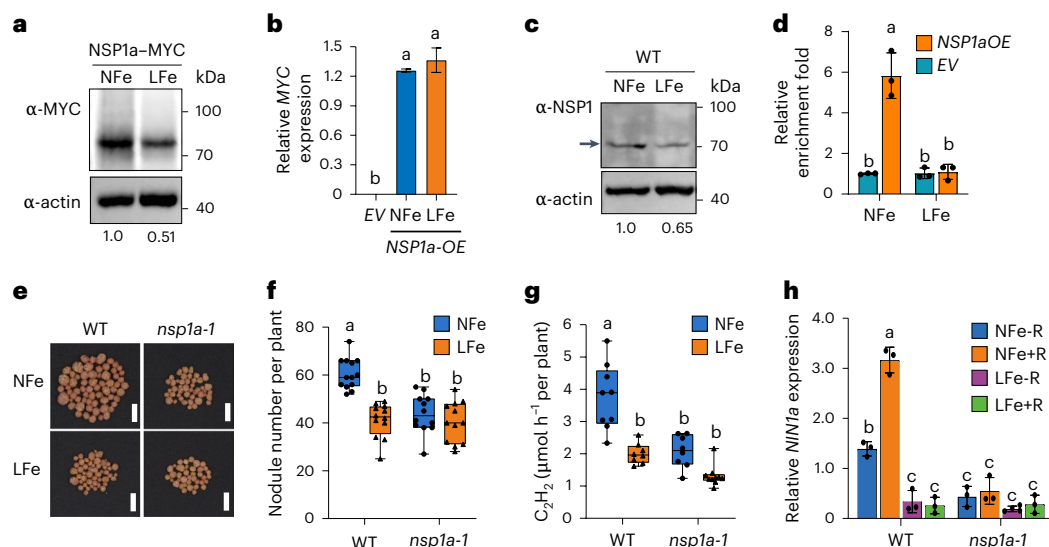


Fig. 2 | NSP1a is involved in low-Fe-mediated inhibition of nodulation.

a, NSP1a-4×MYC abundance in Wm82 hairy roots carrying a *UBI_{pro}::NSP1a-4×MYC* cassette under normal (NFe) and low-iron (LFe) conditions. The numbers at the bottom denote the relative intensities of NSP1a-MYC normalized to the sample under normal iron condition. **b**, NSP1a-MYC transcript levels in Wm82 hairy roots used in **a**; values are means ± s.e.m. of 3 technical repetitions. 'a' represents no significant difference between NFe and LFe based on one-way ANOVA ($P < 0.05$). **c**, Western blot of native NSP1a abundance in wild-type Wm82 roots with GFP-tagged rhizobia under NFe and LFe conditions at 5 DAI. The numbers at the bottom denote the relative intensities of NSP1a normalized to the sample under normal iron condition. The arrow represents NSP1. **d**, ChIP-qPCR assay of NSP1a-4×MYC accumulation at the *NIN1a* promoter. Data are means ± s.d.; $n = 3$. EV, empty-vector negative control. **e**, Nodule phenotypes of

Wm82 and *nsp1a-1* under low-iron conditions. Scale bars, 5 mm. **f**, Effect of iron concentration on nodulation of *nsp1a-1* mutant. Nodules were counted at 21 DAI. Data are means ± s.d.; $n = 13, 12, 12, 12$ (for WT + NFe, WT + LFe, *nsp1a-1* + NFe and *nsp1a-1* + LFe, respectively). **g**, Effect of iron on nitrogenase activity of *nsp1a-1* mutant. Nitrogenase activity was assayed at 21 DAI. Data are means ± s.d.; $n = 9, 8, 8, 10$ (for WT + NFe, WT + LFe, *nsp1a-1* + NFe and *nsp1a-1* + LFe, respectively). **h**, Expression of *NIN1a* in WT and *nsp1a-1* under different iron conditions at 1 DAI. Data are means ± s.d.; $n = 3$. For boxplots (**f**, **g**), whiskers indicate minimum and maximum values, the line indicates the median, and the box boundaries indicate the upper (25th percentile) and lower (75th percentile) quartiles. Different lowercase letters above plots (**d**, **f**–**h**) indicate significant differences determined using two-way ANOVA ($P < 0.05$). The experiments were performed 3 times with similar results.

NSP1a modulates low-iron-mediated inhibition of nodulation

NSP1a is a transcriptional activator of *NIN1a* in soybean nodulation^{11,14}. Because *NSP1a* messenger RNA level upon inoculation was apparently unresponsive to iron supply (Extended Data Fig. 1k), we hypothesized that iron limitation reduces the transcriptional activity of NSP1a on the *NIN1a* promoter. To test this, we first analysed whether different iron regimes affect NSP1a stability by expressing a MYC-tagged NSP1a fusion protein from the constitutive and ubiquitous *UBI1* promoter (*UBI_{pro}::NSP1a-4×MYC*) in transgenic soybean hairy roots grown under normal (10 μM Fe citrate) or iron-deficient (200 μM AA-DP/2,2'-bipyridine) conditions³⁶. NSP1a-4×MYC abundance was downregulated under iron-deficient conditions despite its transcript remaining stable between iron conditions (Fig. 2a,b), suggesting that iron promotes NSP1a accumulation. To confirm this, we assayed native NSP1a abundance in root extracts from plants treated with 10 μM Fe citrate or 200 μM AA-DP using NSP1a native antibody (specificity of antibody is shown in Extended Data Fig. 2a) and detected a downregulation of NSP1a under iron deficiency (Fig. 2c), confirming that iron can stabilize NSP1a in planta.

Next, we examined the influence of iron on NSP1a-4×MYC enrichment at the *NIN1a* promoter by chromatin immunoprecipitation–quantitative polymerase chain reaction (ChIP–qPCR). Strong enrichment was observed in the –1,748 bp and –2,183 bp region of the *NIN1a* promoter in plants grown under normal conditions (Fig. 2d), consistent with a previous report⁶⁷. NSP1a-4×MYC association at the *NIN1a* promoter was sharply reduced under iron deficiency (Fig. 2d), in agreement with *NIN1a* downregulation under iron limitation (Fig. 1h). To further elucidate the function of NSP1a under limited-iron conditions, we generated an *nsp1a* knockout mutant (Extended Data Fig. 2b–d) and characterized nodulation (Fig. 2e–g). The inhibitory effects of iron limitation on nodulation and nitrogenase activity were

nearly completely lost in the *nsp1a-1* mutant (Fig. 2e–g), and *NIN1a* induction by rhizobia was dramatically inhibited (Fig. 2h). Together, these data implicate NSP1a as a factor in transducing low iron to nodulation. The induction and stabilization of NSP1a under sufficient iron subsequently activates *NIN1a*, thereby promoting nodulation.

BTSa physically interacts with NSP1a

The pronounced nodulation phenotype of *nsp1a-1* under low-iron conditions suggests that NSP1a functions to integrate iron availability with nodulation-signalling pathways. To identify putative iron-signalling components, we performed a yeast two-hybrid (Y2H) assay to capture NSP1a-interacting proteins (Supplementary Table 1). Among the detected candidates, Glyma.05g237500 contains metal-ion-binding and DNA-binding domains and is predicted to be an E3 ubiquitin ligase, which is conserved in plants (Extended Data Fig. 3a). Its *Arabidopsis* orthologue BRUTUS (AtBTS) is an iron sensor that negatively regulates responses to iron deficiency⁵⁷. We therefore assigned Glyma.05g237500 as BTSa on the basis of its phylogenetic relationship with AtBTS (Extended Data Fig. 3a). This led us to wonder whether BTSa potentially integrates iron and nodulation cues in soybean. To test this, we first validated the NSP1a–BTSa interaction. Full-length BTSa can interact with NSP1a in a yeast two-hybrid assay (Fig. 3a). This interaction occurs in planta in the nucleus and BTSa-3×HA could be co-immunoprecipitated with NSP1a-GFP (Fig. 3b,c), thus confirming that BTSa is indeed an NSP1a-interacting partner. To dissect which domain(s) mediate(s) the interaction of BTSa with NSP1a, we split BTSa into an N-terminal fragment (BTSa-N^{ter}) that includes its three haemerythrin (HHE) cation-binding domains, and a C-terminal fragment (BTSa-C^{ter}) containing a CHY-type zinc-finger domain, a Rubredoxin-type fold and RING zinc-finger domain (Fig. 3d). In yeast, BTSa-C^{ter}, but not BTSa-N^{ter}, could interact with NSP1a (Fig. 3a), indicating the CHY-type/RING zinc-finger-containing domains are sufficient for this interaction.

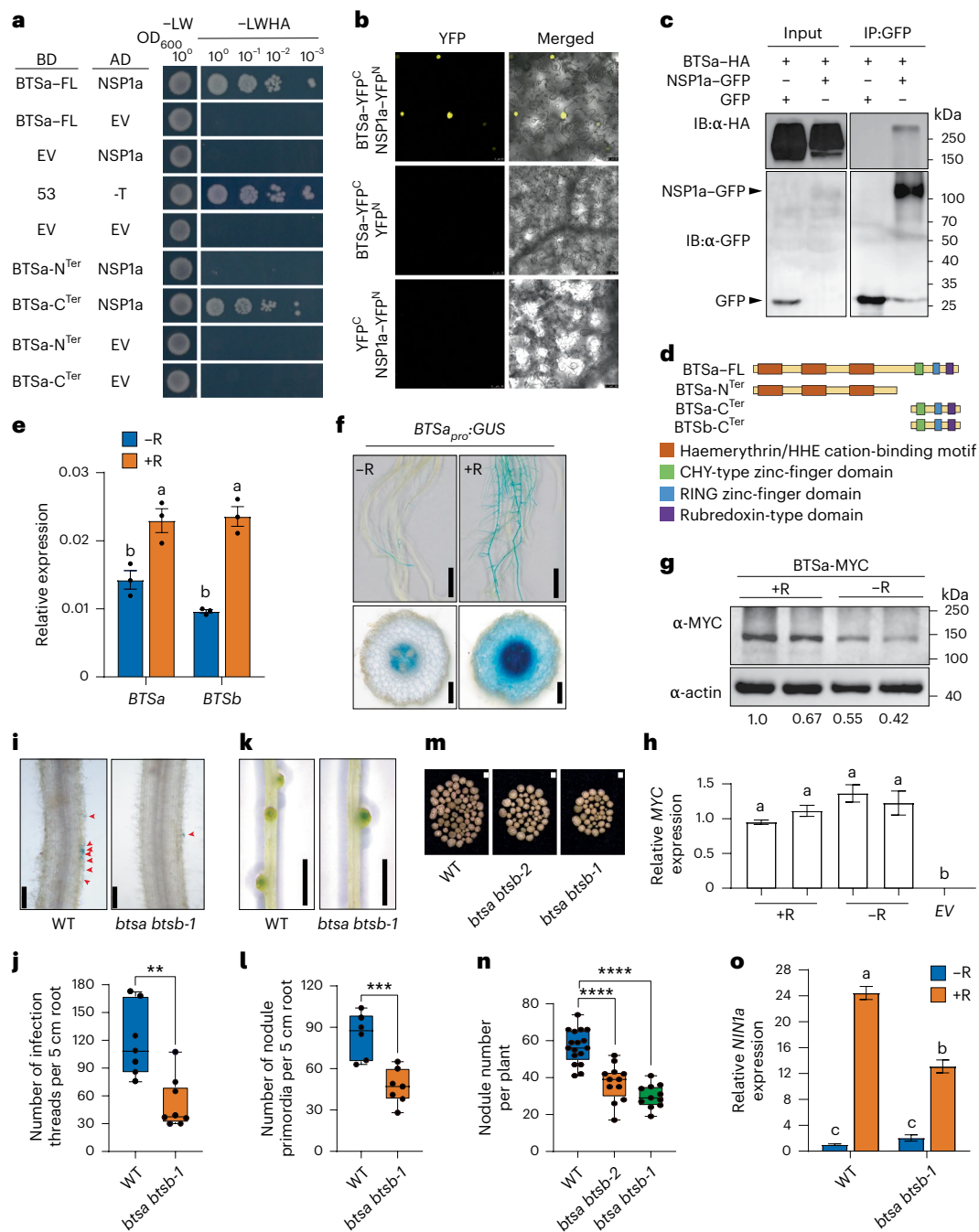


Fig. 3 | BTSa interacts with NSP1a to promote soybean nodulation. **a–c**, NSP1a interacts with BTSa variants in yeast two-hybrid (**a**), BiFC (**b**) and Co-IP (**c**) in *N. benthamiana* leaves. Scale bars, 25 μm. **d**, Domain structures of the full-length and BTS truncations. –LW: Leu and Trp dropout; –LWHA: Leu, Trp, His and Ade dropout. **e**, *BTS* expression in Wm82 roots inoculated (+R) for rhizobia) or uninoculated (–R) with rhizobia. Values are means ± s.e.m. of 3 technical repetitions. The experiments were repeated 3 times with similar results. **f**, GUS staining in transgenic roots expressing *BTSa_{pro}:GUS* at 1 DAI. Scale bars, 10 mm (top) and 100 μm (bottom). **g**, BTSa-4×MYC levels in transgenic roots expressing *UBI_{pro}:BTSa-4×MYC* at 1 DAI; the numbers at the bottom denote the relative intensities of BTSa-MYC normalized to inoculated sample 1. **h**, *BTSa-MYC* transcript levels in hairy roots in **g**. Values are means ± s.e.m. of 3 technical repetitions. **i**, **j**, Representative images (**i**) and number (**j**) of infection threads in Wm82 and *btsa btsb-1* at 3 DAI. The red arrowheads indicate infection threads.

Scale bars, 200 μm. Data are means ± s.d.; *n* = 7 (WT) and 8 (*btsa btsb-1*). **k**, **l**, Representative images (**k**) and number (**l**) of nodule primordia at 6 DAI. Scale bars, 1 mm. Data are means ± s.d.; *n* = 6 (WT) and 7 (*btsa btsb-1*). **m**, **n**, Nodule phenotypes (**m**) and nodule number (**n**) at 21 DAI. Scale bars, 1 mm. Data are means ± s.d.; *n* = 17 (WT), 11 (*btsa btsb-1*), 12 (*btsa btsb-2*). **o**, *NIN1a* expression in roots at 1 DAI. Values are means ± s.e.m. of 3 technical repetitions. The experiments were repeated 3 times with similar results. Different lowercase letters above plots indicate significant differences determined using two-way ANOVA (*P* < 0.05) (**e**, **o**) or one-way ANOVA (*P* < 0.05) (**h**). For boxplots (**j**, **l**, **n**), whiskers indicate the minimum and maximum values, centre lines indicate medians, and box boundaries indicate the upper (25th percentile) and lower (75th percentile) quartiles. Significant differences were determined using two-sided Student's *t*-test; ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001.

Phylogenetic analyses indicate that soybean has four paralogous BTS proteins, of which BTSa and BTSb are highly homologous with 97.2% sequence similarity (Extended Data Fig. 3a,b). Yeast two-hybrid assay showed that BTSb can also interact with NSP1a through its C terminus (Extended Data Fig. 3c). Together, these data suggest that both BTSa and BTSb can physically interact with NSP1a and may function in soybean nodulation.

BTSa and BTSb positively regulate soybean nodulation

To investigate the function of BTSa and BTSb in soybean nodulation, we first assayed their expression upon inoculation with rhizobia. Both *BTSa* and *BTSb* are induced by rhizobia and are highly expressed in nodules (Fig. 3e and Extended Data Fig. 4a,b). *BTSa_{pro}::GUS* activity was restricted to the vasculature of uninoculated roots and was highly induced in whole rhizobia-infected roots (Fig. 3f and Extended Data Fig. 4c). *BTSa_{pro}::GUS* was highly expressed in the cortex and vascular bundles as the nodules developed and matured (Extended Data Fig. 4d). To test whether BTSa abundance is responsive to rhizobia, we transformed *UBI_{pro}::BTSa-4×MYC* into hairy roots with or without rhizobial inoculation. *BTSa-4×MYC* was more strongly expressed in inoculated roots vs uninoculated controls (Fig. 3g), while transgene *BTSa* transcripts were comparable between inoculated and uninoculated hairy roots (Fig. 3h). These results suggest that *BTSa* expression is responsive to rhizobia at both the transcript and protein levels.

We next generated two independent *btsa btsb* double-knockout mutants (assigned *btsa btsb-1* and *btsa btsb-2*) by clustered regularly interspaced short palindromic repeats–CRISPR associated 9 (CRISPR–Cas9) and evaluated their nodulation performance. Both mutant alleles have premature stop codons, express truncated BTSa and BTSb proteins containing partial HHE domains, and show no evidence of off-target edits to the corresponding target sites for *BTSa* and *BTSd* (Extended Data Fig. 4e–g). The number of infection threads, nodule primordia and nodules in both *btsa btsb* double mutants was markedly lower than in wild-type (WT) Wm82 controls (Fig. 3i–n), suggesting a positive role in soybean nodulation. Consistent with the impaired-nodulation phenotypes, *NIN1a* induction by rhizobia in the wild type was strongly attenuated in the *btsa btsb-1* mutant (Fig. 3o). Taken together, these data suggest that BTSa and BTSb are required for nodulation by enhancing pro-nodulation signalling activity.

To investigate whether BTSa and BTSb have distinct roles in nodulation, we expressed *BTSa* or *BTSb* under the control of the ubiquitin promoter or their native promoter in the *btsa btsb* mutant. The expression of *BTSa* or *BTSb* in the *btsa btsb* mutant completely restored its nodulation phenotype (Extended Data Fig. 5a–c). In addition, we generated single *btsa* and *btsb* mutants, and nodule numbers and nitrogenase activity were comparable to those in the wild type (Extended Data Fig. 5d–k). These results indicate that BTSa and BTSb have similar roles in regulating nodulation.

BTSa acts as an iron sensor during nodulation

Since AtBTS has iron-sensing functionality in *Arabidopsis*^{54,57}, we therefore sought to determine whether and how soybean BTSa responds to iron in the context of nodulation. *BTSa* was highly induced by iron deficiency in soybean roots (Fig. 4a). Transgenic hairy roots expressing *BTSa_{pro}::GUS* indicated that *BTSa* promoter activity was upregulated in vascular tissues and extended into the cortical cells under low-iron conditions (Fig. 4b), confirming that *BTSa* is responsive to iron deficiency.

BTSa has three HHE motifs (HHE, iron-binding motif) and a RING zinc-finger domain (Extended Data Fig. 6a). To test iron-binding capability of the HHE motifs, we expressed a full-length recombinant BTSa or a truncated form containing only the three HHE domains fused to maltose-binding protein (MBP) in *E. coli* (Extended Data Fig. 6b) and measured iron concentrations^{56,57} associated with them (Fig. 4c). BTSa can directly bind to iron. Notably, BTSa abundance increased

dramatically in planta in the presence of increasing concentrations of iron (Fig. 4d–f). In addition, BTSa abundance decreased in the presence of deferiprone (an Fe chelator)⁶⁸ but increased in the presence of ferric ammonium citrate when expressed in *N. bethamiana* leaves (Extended Data Fig. 6c). These results suggest that iron stabilizes BTSa, which is in contrast with *Arabidopsis* BTS being stabilized by iron deficiency⁵⁷.

To test whether BTSa and BTSb contribute to soybean's response to iron deficiency, we assayed iron-responsive marker genes including *FRO2* and *IRT1* that herald early responses to iron deficiency in roots. *FRO2* and *IRT1* were sharply induced in the wild type by iron deficiency, but their induction in the *btsa btsb-1* mutant was markedly decreased (Fig. 4g). To test whether BTSa and BTSb have similar roles in responses to iron deficiency, we analysed *FRO2* and *IRT1* expression in the root of *btsa* and *btsb* single mutants. *FRO2* and *IRT1* were reduced in the *btsb* mutant, but were comparable to the wild type in the *btsa* mutant (Fig. 4g), suggesting that BTSb plays a primary role in regulating iron uptake.

We next investigated potential roles for BTSa and BTSb in nodulation under low-iron conditions. In contrast to the wild type, in which there was a substantial decrease in nodulation under iron deficiency, nodule numbers and nitrogenase activity remained unchanged in both *btsa btsb* double mutants under low-iron conditions (Fig. 4h and Extended Data Fig. 6d,e). Consistent with this, *NIN1a* induction by rhizobia was greatly attenuated in the wild type under low-iron conditions but was unchanged in *btsa btsb-1* (Fig. 4i). This suggests that BTSa and BTSb are essential for iron-mediated nodulation. Collectively, these results suggest that BTSa and BTSb together (or alone) function as potential iron sensors that modulate nodulation in response to iron status.

BTS stabilizes and enhances NSP1a transcriptional activity

Since BTSa and BTSb physically interact with NSP1a, we wondered whether these proteins transduce iron signals to trigger nodulation by NSP1a. To test this, we first analysed whether BTS affects *NSP1a* expression. *NSP1a* induction after inoculation was unaffected in the *btsa btsb-1* mutant under normal conditions but decreased under low-iron conditions (Extended Data Fig. 7a). To elucidate whether and how BTSa affects NSP1a protein stability, we first assayed NSP1a degradation in a cell-free system. Recombinant NSP1a–GST was incubated with total-protein extracts from *N. bethamiana* leaf cells transiently expressing *UBI_{pro}::BTSa-3×HA* or an empty vector. NSP1a–GST was degraded much more slowly when incubated with protein extracts containing *BTSa-3×HA* compared with those of empty vector (Extended Data Fig. 7b,c), suggesting that BTSa enhances the stability of NSP1a–GST in vitro. Second, we assessed the effect of BTSa on NSP1a stability in planta. We co-expressed *UBI_{pro}::NSP1a-4×MYC* with *UBI_{pro}::BTSa-3×HA* or the empty vector in *N. bethamiana* leaves. NSP1a–4×MYC accumulated more in leaves co-expressing *UBI_{pro}::BTSa-3×HA* than those of the empty vector (Extended Data Fig. 7d,e). Third, we transformed *UBI_{pro}::NSP1a-4×MYC* into wild-type Wm82 and *btsa btsb-1* hairy roots. NSP1a–4×MYC abundance in *btsa btsb-1* was lower than that in the wild-type control (Fig. 4j), while transgene *NSP1a* transcript levels in *btsa btsb-1* were indistinguishable from those in the wild-type control (Fig. 4k). Finally, we assayed native NSP1a abundance in roots of wild type and both *btsa btsb* mutants. NSP1a in *btsa btsb* mutants decreased compared with the wild type under normal conditions and decreased even further under low-iron conditions (Fig. 4l), indicating that BTSa and BTSb are required for the stability of NSP1a.

To understand whether BTSa and BTSb affect the DNA-binding activity of NSP1a, we performed ChIP assays to verify NSP1a enrichment at the *NIN1a* promoter in hairy roots expressing *UBI_{pro}::NSP1a-4×MYC* in both wild type and *btsa btsb-1* mutant backgrounds. NSP1a–MYC enrichment was markedly reduced in the *btsa btsb-1* background compared with the wild-type control (Fig. 4m). These data indicate that

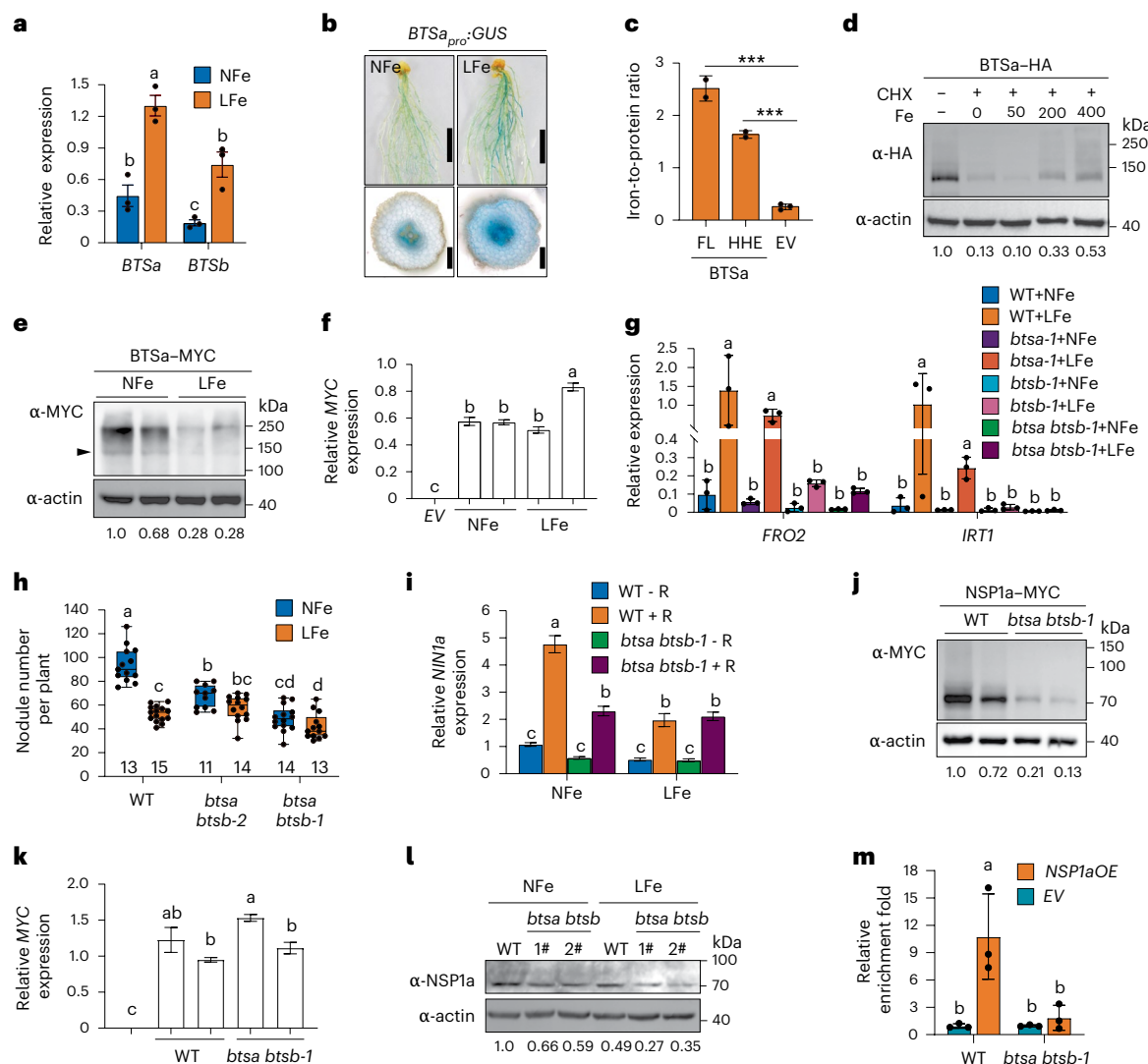


Fig. 4 | BTSa responds to iron stress to regulate NSP1 stability and transcriptional activity. **a**, *BTS* expression in response to normal and low iron. Data are means $\pm$ s.d.; $n = 3$. **b**, GUS staining in roots expressing *BTSa_{pro}::GUS*. Scale bars, 10 mm (top) and 100 μ m (bottom). **c**, Quantification of iron content of *BTSa* variants. FL, full-length MBP-*BTSa*; HHE, MBP fused with three HHE domains; EV, MBP protein. Data are means $\pm$ s.d.; $n = 2, 2, 3$ (for FL, HHE, EV). Significant differences were determined using two-sided Student's *t*-test ($***P < 0.001$). **d**, In vivo stability of *BTSa*-3 $\times$ HA in *N. benthamiana* leaves. The numbers at the bottom denote the relative intensities of *BTSa*-HA normalized to lane 1. CHX, cycloheximide. **e**, *BTSa*-4 $\times$ MYC abundance in transgenic roots under NFe and LFe conditions. The numbers at the bottom denote the relative intensities of *BTSa*-MYC normalized to lane 1. The arrowhead indicates *BTSa*-MYC protein. **f**, Exogenous *BTSa*-MYC transcript levels in **e**. Values are means $\pm$ s.e.m. of 3 technical repetitions. **g**, *FRO2* and *IRT1* expression in the roots. Data are

means $\pm$ s.d.; $n = 3$. **h**, Iron limitation inhibits *btsa btsb* nodulation. Numbers under boxes represent the number of each sample, and data are means $\pm$ s.d. **i**, *NIN1a* expression in the roots. Data are means $\pm$ s.e.; $n = 3$. **j**, *NSP1a*-4 $\times$ MYC abundance in roots expressing *UBI_{pro}::NSP1a*-4 $\times$ MYC. The numbers at the bottom denote the relative intensities of *NSP1a*-MYC normalized to the intensity in WT-background sample 1. **k**, *NSP1a*-MYC transcript levels in roots used in **j**. Values are means $\pm$ s.e.m. of 3 technical repetitions. **l**, *NSP1a* abundance in roots under NFe and LFe treatments at 7 DAI. The numbers at the bottom denote the relative intensities of native *NSP1a* normalized to WT under NFe condition. **m**, ChIP-qPCR of *NSP1a*-4 $\times$ MYC accumulation at the *NIN1a* promoter in transgenic roots. Data are means $\pm$ s.d.; $n = 3$. Different lowercase letters above plots indicate significant differences determined using two-way ANOVA ($P < 0.05$) (**a**, **g**–**i**, **m**) or one-way ANOVA ($P < 0.05$) (**f**, **k**).

BTSa and *BTSb* are crucial for protein stability and the transcriptional activation capacity of *NSP1a*.

BTSa monoubiquitinates *NSP1a*

Because *BTSa* contains a RING-finger domain (Extended Data Fig. 6a), which is responsible for transferring ubiquitin to a substrate protein in E3 ubiquitin–ligase complexes⁶⁹, we wondered whether it ubiquitinates *NSP1a* substrate. First, we tested whether *BTSa* has E3-ligase activity. *BTSa*^{RING}-MBP recombinant protein, when in the presence of a human E1 activating enzyme (hUBE1), an E2 conjugating enzyme (hUBE2) and ubiquitin (Ub-3 $\times$ HA), resulted in auto-ubiquitination (Extended Data

Fig. 8a,b), indicating that *BTSa*^{RING} has functional E3-ligase activity. Second, we tested whether *BTSa* ubiquitinates *NSP1a* in planta. We transiently co-expressed *35S_{pro}::NSP1a*-3 $\times$ FLAG, *35S_{pro}::UBQ*-6 $\times$ MYC and *UBI_{pro}::BTSa*-3 $\times$ HA or the *UBI_{pro}::3 $\times$ HA* empty vector in *N. benthamiana* leaves. Total proteins were immunoprecipitated with anti-MYC beads, and then treated with GST-*USP2*-cc deubiquitinase or heat-inactivated *USP2*. A clear *NSP1a*-FLAG band was detected in the sample co-expressed with *BTSa*-3 $\times$ HA compared with the 3 $\times$ HA empty vector, and importantly, this band disappeared upon GST-*USP2*-cc treatment (Fig. 5a). This single band is not a polyubiquitination phenomenon but a hallmark of a monoubiquitination modification^{70,71}. This

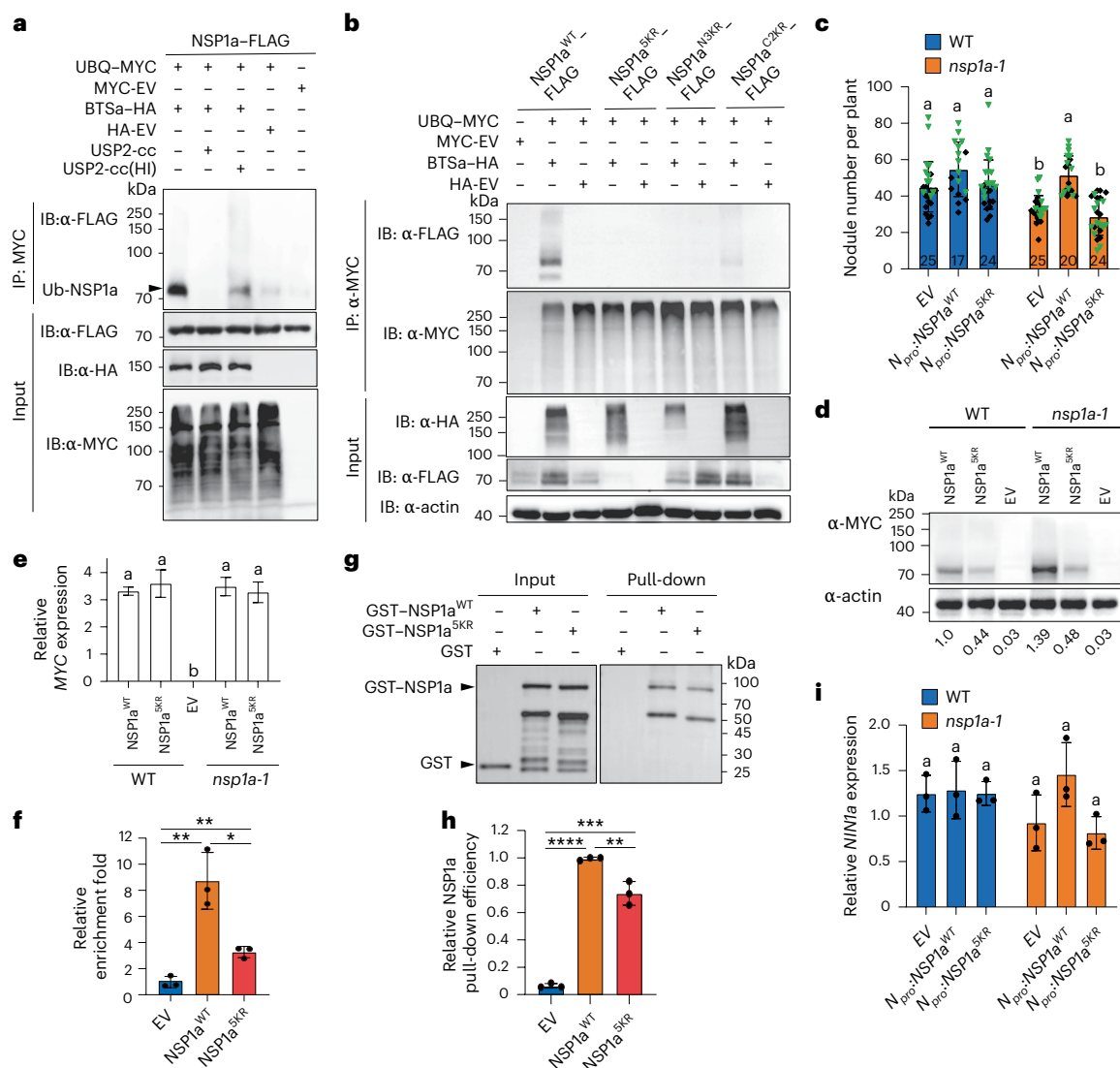


Fig. 5 | BTSA promotes NSP1a stability by monoubiquitination modification.

a, In vivo ubiquitination of NSP1a by BTSA in *N. benthamiana*. Anti-MYC agarose was used for immunoprecipitation and the ubiquitinated complex was treated with or without GST-UBP2-cc. Heat-inactivated USP2 (USP2-cc-HI) was used as a control. **b**, N-terminal three lysine residues are required for NSP1a monoubiquitination. *35S_{pro}*:UBQ-6×MYC, *UBI_{pro}*:BTSA-3×HA and the indicated FLAG-tagged NSP1a mutants were transiently expressed in *N. benthamiana* leaves. **c**, Nodule number of Wm82 and *nsp1a-1* hairy roots transformed with NSP1a_{pro}:NSP1a^{WT}-4×MYC (NSP1a^{WT}) or NSP1a_{pro}:NSP1a^{SKR}-4×MYC (NSP1a^{SKR}). Datapoints with different colours represent different biological repetitions and numbers in the box represent the number of samples. Different lowercase letters above plots indicate significant differences determined using two-way ANOVA ($P < 0.05$). **d**, Protein levels of NSP1a at 1 DAI in Wm82 and *nsp1a-1* roots carrying NSP1a_{pro}:NSP1a^{WT}-4×MYC (NSP1a^{WT}) or NSP1a_{pro}:NSP1a^{SKR}-4×MYC (NSP1a^{SKR}) constructs. The numbers at the bottom denote the relative intensities

of NSP1a-MYC normalized to line 1. **e**, Exogenous NSP1a-MYC transcript levels in Wm82 roots used in **d**. Values are means $\pm$ s.e.m. of 3 technical repetitions. Different lowercase letters above plots indicate significant differences determined using one-way ANOVA ($P < 0.05$). **f**, ChIP-qPCR of NSP1a-4×MYC enrichment at the *NIN1a* promoter. Data are means $\pm$ s.d.; $n = 3$. Significant differences were determined using two-sided Student's *t*-test ($*P < 0.05$, $**P < 0.01$). **g**, DNA-protein pull-down assay to probe the effects of NSP1a-GST ubiquitination on its binding to the *NIN1a* promoter. **h**, Quantification of relative pull-down efficiency in **g** using ImageJ. Data are means $\pm$ s.d.; $n = 3$. Significant differences were determined using two-sided Student's *t*-test ($**P < 0.01$, $***P < 0.001$, $****P < 0.0001$). **i**, *NIN1a* expression in Wm82 and *nsp1a-1* hairy roots at 1 DAI. Data are means $\pm$ s.d.; $n = 3$. Different lowercase letters above plots indicate significant differences determined using two-way ANOVA ($P < 0.05$). The experiments were repeated 3 times with similar results.

result reveals that BTSA is responsible for post-translational modification of NSP1a by monoubiquitination.

To identify BTSA-mediated NSP1a monoubiquitination sites, we performed liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis of in vivo ubiquitinated NSP1a. Five ubiquitinated lysine residues were detected (Extended Data Fig. 8c–g). Among them, three (K133, K139 and K143) are in the low-complexity region of the N-terminal end (N^{ter}) and two (K398 and K468) are in the C-terminal end (C^{ter}) (Extended Data Fig. 8c). To confirm the monoubiquitination sites, we mutated all three lysine residues in N^{ter} (NSP1a^{N3KR}), both lysine

residues in C^{ter} (NSP1a^{C2KR}) or all five lysine residues (NSP1a^{SKR}), and tested their ubiquitination activity in vivo. NSP1a^{C2KR} compromised NSP1a monoubiquitination, while NSP1a^{N3KR} and NSP1a^{SKR} completely blocked NSP1a monoubiquitination (Fig. 5b), suggesting that the N^{ter} lysine residues are required for NSP1a monoubiquitination by BTSA. In addition, we found that NSP1b, the closest homologue of NSP1a, but not their coactivators NSP2a/b, contains the three conserved lysine residues for monoubiquitination by BTSA (Extended Data Fig. 9a). In vivo ubiquitination assays showed that NSP1b, but not NSP2a, was monoubiquitinated by BTSA (Extended Data Fig. 9b,c), suggesting

that monoubiquitination of NSP1 by BTSa is substrate specific. To test whether NSP1a monoubiquitination affects its function in nodulation, we transformed *NSP1a*^{WT}-4×MYC and *NSP1a*^{SKR}-4×MYC driven by its native promoter into wild-type and *nsp1a-1* hairy roots. Neither expression of *NSP1a*^{WT}-4×MYC nor of *NSP1a*^{SKR}-4×MYC affected nodule numbers in the wild type. By contrast, *NSP1a*^{WT}-4×MYC in *nsp1a-1* restored nodule number to wild-type levels, but *NSP1a*^{SKR}-4×MYC did not complement the nodulation defect (Fig. 5c and Extended Data Fig. 9d). These results demonstrate that NSP1a monoubiquitination plays a key role in promoting nodulation.

Monoubiquitination affects NSP1a stability and activity

Monoubiquitination affects protein stability and transcriptional activity^{72,73}. We next tested how monoubiquitination affects NSP1a function. We first probed *NSP1a*^{WT}-4×MYC and *NSP1a*^{SKR}-4×MYC levels in both wild-type and *nsp1a-1* hairy roots. *NSP1a*^{SKR}-4×MYC was less abundant in both backgrounds, although transgene transcript levels were comparable (Fig. 5d,e). Furthermore, *NSP1a*^{SKR}-4×MYC enrichment at the *NIN1a* promoter was remarkably reduced (Fig. 5f), and the DNA-binding ability of *NSP1a*^{SKR}-4×MYC was the same as that of wild-type NSP1a under iron deficiency (Fig. 2d). DNA–protein pull-down assay confirmed that *NSP1a*^{SKR}-GST had reduced binding activity to the *NIN1a* promoter (Fig. 5g,h) and failed to activate *NIN1a* expression (Fig. 5i). Collectively, these results suggest that the monoubiquitination of NSP1a by BTSa is necessary for protein accumulation and proper transcriptional activation activity of NSP1a.

Monoubiquitination of NSP1 by BTS is conserved in legumes

To confirm whether NSP1 monoubiquitination is conserved, we first aligned NSP1 sequences from soybean with those from the Papilionoideae legumes *Medicago* (*M.*) *truncatula* and *Lotus* (*L.*) *japonicus*. Three lysine residues (K133, K139 and K143) in the N-terminal end are conserved in soybean, *M. truncatula* and *L. japonicus*; K398 is conserved in soybean and *M. truncatula*, while K468 is conserved in soybean and *L. japonicus* (Extended Data Fig. 9a), suggesting that NSP1 proteins in *M. truncatula* and *L. japonicus* are probably monoubiquitinated by BTS E3-ligase activity. Yeast two-hybrid and co-immunoprecipitation (Co-IP) experiments revealed that MtNSP1 and LjNSP1 can indeed interact with MtBTS and LjBTS, respectively (Fig. 6a,b). In vivo ubiquitination assays showed that both FLAG-tagged MtNSP1 and LjNSP1 were monoubiquitinated by their MtBTS and LjBTS partners, and that mutation of the three conserved K residues in MtNSP1 and LjNSP1 completely blocked monoubiquitination by BTS (Fig. 6c), confirming a conserved role for these lysine residues in N-terminal monoubiquitination of legume NSP1s.

To validate the role of MtNSP1 and LjNSP1 monoubiquitination in nodulation, we transformed MtNSP1^{WT}-4×MYC, LjNSP1^{WT}-4×MYC and respective mutant forms MtNSP1^{SKR}-4×MYC and LjNSP1^{SKR}-4×MYC into wild type and *nsp1a-1* hairy roots with constructs driven by the *UBI* promoter. MtNSP1-4×MYC and LjNSP1-4×MYC completely rescued the nodulation defect of *nsp1a-1* to the wild-type level, while MtNSP1a^{SKR}-4×MYC and LjNSP1a^{SKR}-4×MYC did not complement the *nsp1a-1* mutant (Fig. 6d and Extended Data Fig. 10a). These data suggest that BTS-mediated monoubiquitination of NSP1 is conserved in legumes. To confirm the legume-specific role of NSP1 monoubiquitination by BTSa, we performed a monoubiquitination assay of NSP1 using *Arabidopsis* AtBTS. FLAG-tagged NSP1a was not monoubiquitinated by AtBTSa (Extended Data Fig. 10b), and a Co-IP assay revealed no detectable interaction between NSP1 and AtBTS (Extended Data Fig. 10c). Collectively, these results demonstrated the legume-specific monoubiquitination of NSP1 by BTSa, which is essential for integrating iron signals and promoting legume nodulation.

Discussion

Iron has long been appreciated as crucial for symbiotic nitrogen fixation in legumes because nitrogenase and leghaemoglobin require vast

amounts of iron^{41,42,45,74,75}. Nodulation is a prerequisite for the establishment of symbiotic nitrogen fixation. As both the host and partner need iron, we propose that sensing iron status and coordinating iron availability by the host during nodulation under varying iron conditions must be an important strategy for the successful establishment of symbiosis. In this work, we provide evidence that iron availability plays a decisive role in the successful nodulation of soybean. We have identified BTSa as an iron sensor that integrates iron and nodulation signalling pathways upon perception of iron status. Our data support a model in which BTSa senses and transduces the early iron signal to NSP1, which elicits nodulation signalling in inoculated roots to promote nodulation.

We observed that symbiotic nodulation exhibits a high dependence on iron. The absence of iron completely inhibited soybean nodule initiation even under low-nitrogen and sufficient light conditions. The existing theory is that nitrogen and light are two strong drivers of legume nodulation and symbiotic nitrogen fixation^{1,30–33}. Our observation that iron deficiency blocks soybean nodulation under the conditions of low nitrogen and sufficient light suggests that iron may be as strong a driver as nitrogen and light availability, a notion supported by observations in *Lupinus* nodulation^{42–44}. Furthermore, we identified a suitable iron concentration (~40 μM) for soybean nodulation, while excess iron levels (>160 μM) were found to inhibit nodulation due to excessive ROS accumulation and cell toxicity. Different leguminous plants and rhizobia may have different iron requirements. Identification of an ‘iron optimum’ for nodulation may represent a mutual strategy by both plants and rhizobia to adapt to varying levels of iron in natural environments. Altogether, our data suggest that host roots can perceive external variation in iron precisely, to coordinate nodulation signalling pathways, which in turn determines the number of nodules formed and the amount of nitrogen fixed. However, understanding exactly how the host coordinates the establishment of a symbiotic yet energetically demanding relationship with rhizobia based on iron availability during nodulation is still in its infancy.

We saw that the essential nodulation regulators *NIN*s were down-regulated under low-Fe conditions, whereas their transcriptional activators *NSP1* and *NSP2* remained unaffected. Legume NSP1 proteins are central transcription factors in nodulation signalling^{11,12,14}, the induction of which also mediates nitrogen and phosphorus exchange between host and symbionts^{76–78}. It is possible that NSP1 proteins are regulated by nutritional signals to enhance transcriptional activation activity and subsequently promote symbiosis. Indeed, we demonstrate that NSP1a is involved in iron-mediated nodulation. First, we found that NSP1a protein stability strongly correlates with levels of external iron, and NSP1a stability was high in the presence of sufficient iron, with significant decreases under low-iron conditions. Second, we showed that the binding activity of NSP1a to *NIN1a* promoter is affected by iron status, with lower binding activity under low-Fe conditions. Third, and most importantly, the inhibitory effect of low Fe on nodulation is attenuated in *nsp1* mutants. These NSP1a-dependent responses indicate that NSP1a is a key node in perceiving and transducing low external iron signals to nodulation in soybean roots.

How does NSP1a coordinate iron-regulated nodulation? We speculate that there is a factor involved in sensing iron status that transmits the signal to NSP1a. Intriguingly, through the search for NSP1a-interacting proteins, we identified BTSa, a putative orthologue of iron sensor *Arabidopsis* BTS, as an interacting partner. BTSb is highly similar to BTSa and also interacts with NSP1a. Downregulation of *BTSa* and *BTSb* reduces the number of nodules under normal iron conditions⁶³, suggesting important roles for these two proteins in regulating nodulation. *BTSa* transcripts are induced by rhizobia and iron deficiency, while its protein product is also induced by rhizobia and increasing amounts of iron. *btsa btsb* double mutants display decreased nodule number and nitrogenase activity and either *BTSa* or *BTSb* can complement the phenotype of *btsa btsb*. Importantly, the inhibition

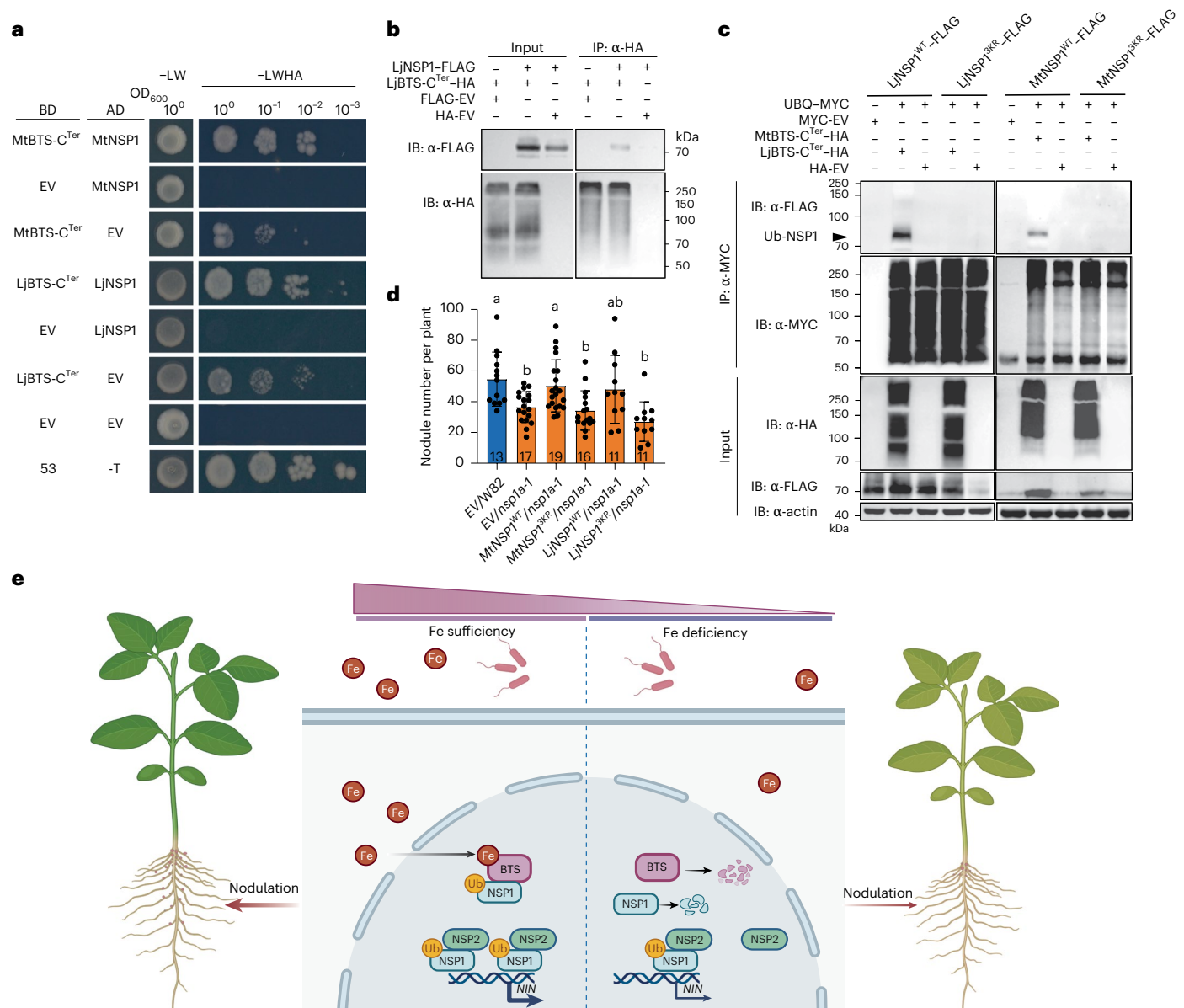


Fig. 6 | Monoubiquitination of NSP1 by BTS is conserved in legumes.

a, Yeast two-hybrid assay to verify the interaction between MtBTS + MtNSP1 and LjBTS + LjNSP1. **b**, Co-IP of the interactions of LjBTS-3×HA with LjNSP1-FLAG. **c**, The three N-terminal lysine residues are required for MtNSP1 and LjNSP monoubiquitination. *35S_{pro}:UBQ-6×MYC*, *UBI_{pro}:LjBTSa-3×HA* or *UBI_{pro}:MtBTSa-3×HA* and respective FLAG-tagged LjNSP1 or MtNSP1 mutants were transiently expressed in tobacco leaves. Anti-MYC agarose was used for immunoprecipitation and immunoprecipitated proteins were probed with anti-HA and anti-FLAG antibodies. **d**, Nodule numbers of transgenic soybean hairy roots in the Wm82 and *nsp1a-1* mutant backgrounds heterologously expressing NSP1 variants. Data are means ± s.d. Numbers in the box represent the number of

samples. Different lowercase letters above plots indicate significant differences determined using one-way ANOVA ($P < 0.05$). **e**, Proposed model for the iron-BTS-NSP1 cascade in soybean nodulation. Under iron-sufficient conditions, BTS is stabilized by binding iron, and then BTS interacts with and stabilizes NSP1 to promote nodulation by enhancing its *NIN* transcriptional activation activity upon monoubiquitination. Under iron deprivation, BTS is unstable and monoubiquitination activity is reduced, leading to a decrease in NSP1 protein stability and the transcriptional activation activity of NSP1, resulting in downregulated *NIN* and consequently reduced nodulation. Model created with BioRender.com. In **a-d**, experiments were repeated 3 times with similar results.

of nodulation by low iron is attenuated in *btsa btsb* double mutants. These results suggest that BTSa/b are strong candidates for perceiving and transducing the iron signal to nodulation, through NSP1a. Further supporting this hypothesis, we demonstrated that BTSa can directly bind iron, thereby stabilizing as iron levels increase to enhance uptake of iron. Iron binding probably induces structural changes in BTSa, thus impeding protein degradation. This observation is different from that of *Arabidopsis* BTS, which is unstable upon iron binding and negatively regulates iron response⁵⁷, suggesting that soybean BTSa/b may have diversified structurally and functionally. In-depth analysis

of the molecular mechanisms controlling BTS stability will enable us to better understand the regulatory network of iron-level responses in plants during nodulation and development processes. Thus, our findings lead to a proposed model in which rhizobia induce BTSa/b under sufficient iron condition, with BTSa/b perceiving an early status of iron by binding to iron, leading to the stabilization of NSP1 and transmitting the iron signal to the nodulation signalling pathway.

Arabidopsis BTS functions as an E3 ligase, which polyubiquitinates and degrades bHLH105/115 to negatively regulate Fe uptake^{56,57}. In soybean, both BTSa/b and NSP1a are positive regulators of nodulation;

it is unlikely that BTSa/b polyubiquitinates and degrades NSP1. Interestingly, we have discovered that soybean BTSa/b, but not AtBTS, physically interact with and monoubiquitinate NSP1a to stabilize NSP1a. We identified three lysine residues in the N terminus of NSP1a as the site for its monoubiquitination. Moreover, our data revealed that rhizobia and iron-induced monoubiquitination of NSP1a by BTSa enhances NSP1a stability and subsequent transcriptional activity on the *NIN1a* promoter, thereby activating nodulation signalling and promoting nodulation. These results suggest that monoubiquitination of soybean BTSa on NSP1 is a neofunctionalization. The observation that the *btsa btsb* double mutant did not completely inhibit *NIN1a* induction by rhizobia and nodulation indicates that other BTSs probably mediate this process. This finding uncovers iron sensing and signal-transduction versatility of BTS across plant taxa to provide advantages in integrating iron status with distinct signalling networks to steer adaptation. Due to the presence of four homologues of BTS in soybean, we cannot rule out the possibility that other soybean BTSs exert their functions by polyubiquitinating and degrading their substrates. Subsequent functional and biochemical analyses of the remaining soybean BTSs will facilitate a comprehensive understanding of the regulatory function of these iron-sensing proteins. BTSs exert their function by transducing an iron signal to downstream effectors, such as IMA peptides. LjIMA1 plays a role in nodule iron uptake and regulates the nitrogen–iron balance⁶³. However, the relationship between BTS and IMA in regulating iron uptake during early nodulation is still unclear. Further research elucidating the interaction between IMA and BTS in soybean will enhance our comprehension of iron-signal sensing and cellular iron homeostasis in the establishment of symbiotic relationships.

Our work highlights that BTSa/b-mediated signalling is at the centre of the early root responses to low iron and rhizobial-derived signals. Notably, our data show that BTSa does not interact with and monoubiquitinate NSP2a, suggesting that BTSa monoubiquitination of NSP1 is substrate specific. In addition, our results showed that BTS-mediated monoubiquitination of NSP1 is conserved in nodulating legumes. The observations that (1) AtBTS does not interact with and monoubiquitinate NSP1a, (2) only NSP1 proteins in nodulating legumes contain three conserved lysine residues at their N terminus for monoubiquitination, and (3) LjRING, the homologue of BTS, positively regulates nodulation⁶⁴, support this conclusion. Our study has uncovered a specific mechanism by which the BTS–NSP1 module facilitates the coupling of iron and nodulation signalling pathways to regulate iron homeostasis and nodulation. This mechanism is essential for nutritional nodulation in legumes to adapt to fluctuations in iron levels in the environment. Altogether, our findings help to explain how iron status is sensed in the early stages of nodulation, why iron limitation inhibits nodulation, and why three lysine residues in the N terminus of NSP1 are conserved in legumes. Because NSP1 is a key regulator of symbioses between hosts and nitrogen-fixing bacteria, as well as beneficial mycorrhizal fungi, our findings will help to improve our understanding of the molecular mechanisms that govern the nutritional regulation of symbiotic relationships. Based on the diversity and geographic distribution of symbiotic nitrogen fixers and iron, the importance of iron nutrition in nodulation establishes it as a potential new target for improving nitrogen-fixation efficiency of leguminous crops and other symbiotic nitrogen fixers via genetic and agronomic approaches in the face of increases in arable-land area becoming saline and alkaline under climate change.

Methods

Plant growth and rhizobium inoculation

Soybean *Glycine max* (L.) Merrill cv. Williams 82 (Wm82) plants were used for gene cloning, gene expression and functional analysis. Soybean seedlings were grown in a growth room (16 h light/8 h dark, 25 °C, 50% relative humidity). Two iron-treatment methods were used:

hydroponic culture and vermiculite culture. For hydroponic systems, chlorine-sterilized soybean seeds were germinated in quartz sand moistened with Low Nitrate (LN) nutrient solution [KNO_3 (150 μM), $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ (120 μM), NH_4NO_3 (40 μM), CaCl_2 (1.2 mM), $(\text{NH}_4)_2\text{SO}_4$ (30 μM), KH_2PO_4 (0.5 mM), MgSO_4 (0.5 mM), MgCl_2 (0.025 mM), MnSO_4 (1.5 μM), ZnSO_4 (1.5 μM), CuSO_4 (0.5 μM), $\text{H}_3\text{B}_2\text{O}_7 \cdot \text{Na}_2\text{O} \cdot 2\text{H}_2\text{O}$ (0.15 μM), CoCl_2 (0.1 μM), $\text{Na}_2\text{B}_4\text{O}_7$ (2.5 μM), K_2SO_4 (0.3 mM), pH 5.7] with different Fe–EDTA (formulated with ferrous sulfate and EDTA-2Na) concentrations, and roots of 7-day-old seedlings were soaked in GUS-tagged *B. diazoefficiens* USDA110 (optical density (OD)_{600 nm} = 1.0) solution for 3 h before transplanting into a hydroponic system. The nutrient solution was ventilated using breather pumps and the nutrient solution was changed every 3 d. The number of infection threads and nodule primordia were counted at 3 and 6 d after inoculation (DAI), respectively. Nodule numbers were recorded at different indicated times after inoculation.

For the vermiculite culture method, iron treatments were conducted as previously described with some modifications³⁶. Briefly, seeds were germinated in distilled water-moistened vermiculite, 5-day-old seedlings were transplanted to vermiculite moistened with Broughton and Dilworth Low Nitrate (0.25 mM nitrate) (BD–LN) nutrient solution [CaCl_2 (0.87 mM), $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$ (0.13 mM), KH_2PO_4 (0.5 mM), $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (0.25 mM), K_2SO_4 (1.0 mM), $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ (1 μM), H_2BO_3 (2 μM), $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (0.5 μM), $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (0.3 μM), $\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$ (0.1 μM) and $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$ (0.1 μM)] containing 10 μM Fe citrate ($\text{C}_6\text{H}_5\text{O}_7\text{Fe}$) or 200 μM AA-DP/2,2'-bipyridine, and the cotyledons were removed at 8 d after germination (DAG). Soybean seedlings were inoculated with 30 ml GUS-tagged *B. diazoefficiens* USDA110 ($\text{OD}_{600} = 0.08$) at 10 DAG. Distilled water and BD–LN solution supplemented with 10 μM Fe citrate or 200 μM AA-DP solution were irrigated alternately during growth. The number of infection threads, nodule primordia and nodules were counted at 3, 5 and 21 DAI, respectively.

Nicotiana benthamiana plants were grown in vermiculite in growth chambers (16 h light/8 h dark, 25 °C, 50% relative humidity) for 5 weeks and used for transient expression of proteins in bimolecular fluorescence complementation (BiFC) and stability experiments.

DAB and NBT staining

Lateral roots treated with different iron concentrations at 1 DAI were used for DAB and NBT staining. For DAB staining, the samples were submerged in 0.5 mg ml^{−1} DAB solution (CSB-K09758AB-2, CUSABIO) for 2 h, followed by washing with 95% (v/v) ethanol to decolour. For NBT staining, samples were submerged in 1 mg ml^{−1} NBT staining solution for 10 min, followed by washing with 95% (v/v) ethanol.

ROS quantification

Lateral roots treated with different iron concentrations at 1 DAI were used for ROS quantification. Fresh root samples (1 g) were ground into a homogenate in 10 ml PBS (pH 7.4) and centrifuged at 3,000 r.p.m. at 4 °C for 20 min. A volume of 10 μl of supernatant was used to determine ROS contents using the Plant ROS ELISA kit (Mibio), as previously described⁷⁹.

Plasmid construction

BTSa and *BTSb* were cloned by PCR from Wm82 nodule complementary (c)DNA. For constructs used in Y2H and BiFC assays, coding sequences were amplified and cloned into pDONR207 by attB × attP reactions and the resulting entry constructs were recombined into pGBKT7-DEST (BD), pGADT7-DEST (AD), pEarlygate201(YFP^N) or pEarlygate202(YFP^C). The *MtNSP1*, *LjNSP1*, *MtBTS-C^{Ter}* (C terminus) and *LjBTS-C^{Ter}* coding sequences were amplified and cloned into pGBKT7 and pGADT7 vectors between *EcoRI* and *BamHI* sites. The NSP1a-related vectors (NSP1a–AD and NSP1a–YFP^N) were previously described⁶⁷.

For the 35S_{pro}:NSP1a–GFP, UBI_{pro}:BTSa–3×HA, UBI_{pro}:LjBTSaC^{Ter}–3×HA, UBI_{pro}:MtBTS-C^{Ter}–3×HA, 35S_{pro}:NSP1a–3×FLAG, 35S_{pro}:LjNSP1a–3×FLAG

and 35S_{pro}:*MtNSP1a*-3×*FLAG* constructs, coding sequences were amplified and cloned into pCambia1300, *UBI_{pro}*:*HA* (pCambia1300) and pTF101-3×*FLAG* vectors using *Xba*I and *Bam*HI. For the *UBI_{pro}*:*BTSa*-4×*MYC* and *UBI_{pro}*:*NSP1a*-4×*MYC* constructs, coding sequences were amplified and cloned into the *UBI_{pro}*:4×*MYC* vector between *Stu*I and *Kpn*I sites. *NSP1* point mutations in the pTF101-3×*FLAG* vectors were generated by site-directed mutagenesis (AAA to AGA or AAG to AGG) with primers listed in Supplementary Table 2.

For the *BTSa_{pro}*:*GUS* reporter construct, the promoter (3,087 bp upstream of the ATG translational start codon) was amplified and cloned into pCambia1391 vector between *Bam*HI and *Eco*RI sites. For the 35S_{pro}:*UBQ*-6×*MYC* construct, the monomeric ubiquitin sequence was amplified using *POLYUBIQUITIN 4 (Glyma.13G117700)* as template and cloned into 35S_{pro}:6×*MYC*-pCambia1300 using *Kpn*I and *Sac*I.

For *NSP1a_{pro}*:*NSP1a*-4×*MYC* and its point-mutation variants, the *UBI* promoter in *UBI_{pro}*:*NSP1a*-4×*MYC* constructs was replaced with the *NSP1a* promoter (3,789 bp upstream of ATG) using *Pme*I and *Hind*III with *NSP1a* primers listed in Supplementary Table 2.

For the *BTSa*-MBP construct and its truncated variants, the coding sequence was amplified and cloned into pMal-c2X vector using *Bam*HI and *Sac*I. The *NSP1a*-GST and USP2-cc-GST constructs, as well as the catalytic domain of mouse deubiquitinase USP2 (NM_001407818) were cloned into pGEX-4T-1 vector using *Bam*HI and *Eco*RI.

Generation of transgenic plants

For CRISPR-Cas9 vector construction, the pYLCRISPR/Cas9-DB vector was used as previously reported⁸⁰ and small guide RNAs were designed using CRISPR-P2.0 software (<http://crispr.hzau.edu.cn/CRISPR2/>)⁸¹. Four sgRNAs targeting both *BTSa* and *BTSb* were selected and integrated into different single-guide DNA expression cassettes and then introduced into the pYLCRISPR/Cas9-DB. CRISPR-Cas9-edited mutants were generated using the *Agrobacterium*-mediated cotyledon-node transformation system following previously described approaches^{32,82}.

Briefly, seeds were kept in a vacuum desiccator with chlorine gas for 14 h and then placed in sterile water overnight. The seed coat was gently removed, and two cotyledon-node explants were generated by making a horizontal slice through the hypocotyl and a vertical slice between the cotyledons. Explants were immersed in the inoculum of *Agrobacterium tumefaciens* EHA101 for 8 h with slow horizontal shaking, and then dried on filter paper and transferred to co-cultivation plates for 4 d at 25 °C in darkness. Following co-cultivation, explants were cultured on shoot-initiation medium, with the hypocotyl and cotyledon nodes embedded in the medium under a 12 h light/12 h dark photoperiod at 25 °C and subcultured biweekly to fresh medium. Explants with buds were transferred to shoot-elongation medium and subcultured biweekly in the same conditions until shoots elongated. Elongated shoots (>3 cm) were transferred to rooting medium and the rooted putative transgenic plants were then transplanted to vermiculite for further growth in a growth room. Putative transgenic plants were then characterized using a combination of methods including leaf painting with 0.2 g l⁻¹ Basta solution to check for integration of the *BAR* gene, followed by DNA sequencing to characterize editing of target and homologous genes.

Nitrogenase activity

Nitrogenase activity of nodules was determined using the acetylene-reduction-activity assay as previously described⁸³. One nodulated root was placed inside a sealed 40 ml vial. Of the air inside the vial, 2 ml was removed and then replaced with 2 ml pure acetylene gas under water. Vials were placed upside down and incubated at 28 °C for 2 h. The acetylene-reduction reaction was stopped by incubating vials at 4 °C for 5 min. One hundred microlitres of the gas in each vial was used for the measurement of ethylene content using a GC-4000A gas chromatograph (Dongxi). Nitrogenase activity was calculated by the amount of ethylene reduced per plant per unit of time.

DNA extraction and identification of transgenes

Plant samples were used for DNA extraction using hexadecyltrimethylammonium bromide (CTAB) as previously described⁸⁴. Briefly, leaves or hairy roots were ground in extraction buffer (100 mM Tris-HCl pH 8.0, 2% (w/v) CTAB, 1.4 M NaCl, 0.2% (v/v) 2-mercaptoethanol, 20 mM EDTA) and incubated at 65 °C for 30 min before DNA was extracted with chloroform-isoamyl alcohol (2:1). After centrifugation (12,000 × g for 10 min), the aqueous phase was transferred to fresh tubes and DNA was precipitated using ethanol (1:2).

The stable-transgenic mutant *btsa btsb* plants were first screened by leaf painting with 0.8% w/v glufosinate solution and glufosinate-resistant plants were confirmed by PCR detection of the *BAR* gene using primers listed in Supplementary Table 2. PCR-positive plants were sequenced with *BTS*-specific primers (Supplementary Table 2).

Transgenic hairy roots transformed with *UBI_{pro}*:*BTSa*-4×*MYC*, *UBI_{pro}*:*NSP1a*-4×*MYC* or the empty vector (with GFP reporter in another cassette) were screened using a portable fluorescence lamp to visualize GFP green fluorescence, and verified by gene expression.

RNA extraction and quantitative real-time PCR

Total RNA was extracted using TRIpure Reagent (Aidlab Biotechnologies). Genomic DNA was removed and cDNAs were produced using a Hifair II 1st Strand cDNA Synthesis SuperMix (Yeasen Biotech) with 0.5 µg RNA. Quantitative RT-PCR (RT-qPCR) was performed using the following steps: each reaction contained 1 µl of 1:10 diluted cDNA with 0.25 µl of each primer (10 µM), 5 µl of Hieff qPCR SYBR Green Master Mix kit (Yeasen Biotech) and water to a final volume of 10 µl. The PCR programme was 95 °C for 30 s, followed by 40 cycles (95 °C for 5 s, 60 °C for 34 s) and the melt-curve stage. Amplicon integrity was confirmed by melt-curve analysis. Relative expression levels were calculated using the 2^{-ΔΔC_t} method. *ELF1b* was used as a reference gene, and gene-specific primers are listed in Supplementary Table 2.

Yeast two-hybrid assay

The *NSP1a* coding sequence was fused with the AD domain in pGADT7-DEST. *BTSa*, *BTSb* and their truncated domains were fused with the BD domain in pGBKT7-DEST. To verify the interactions between *NSP1a* and *BTSa*/*BTSb*, the above constructs were co-transformed into *Saccharomyces cerevisiae* AH109. Clones were grown on selective medium (lacking Trp, Leu, Ade and His) to test protein-protein interaction. Images were taken 3 d after incubation at 30 °C.

BiFC assay

The coding sequences of *NSP1a* were cloned into pEarlygate201-YFP^N and *BTSa* was cloned into pEarlygate202-YFP^C through the Gateway reaction system (Invitrogen). The constructs were transformed into *A. tumefaciens* GV3101 separately and cultured in LB medium. Transformed strains were collected and resuspended in infiltration buffer (10 mM MgCl₂, 200 µM acetosyringone pH 5.6) and incubated for 2–3 h at 30 °C before infiltration. Suspensions (OD₆₀₀ = 0.2) were infiltrated into 5-week-old *N. benthamiana* leaves in different combinations. Two days after infiltration, YFP fluorescence was detected using a Leica confocal laser scanning microscope (Leica Microsystems).

Co-IP assay

The *BTSa*-3×*HA* construct was co-expressed with *NSP1a*-GFP in *N. benthamiana* leaves. Total proteins were extracted in extraction buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 1 mM EDTA, 10% (v/v) glycerol, 1% (v/v) Triton X-100, 1 mM PMSF and 1× protease inhibitor cocktail) and incubated with 8 µl anti-GFP-Trap Agarose (ChromoTek, gta) at 4 °C for 2 h. Beads were washed five times with extraction buffer, and the immunoprecipitated proteins were eluted with SDS loading buffer with boiling for 5 min. Samples were analysed by western blotting with anti-*HA* (Sigma, SAB4300603, 1:2,500) and anti-GFP (AbClonal, AE012, 1:2,500) antibodies.

Soybean hairy-root transformation and *B. diazoefficiens* inoculation assay

Hairy-root transformation with *Agrobacterium rhizogenes* K599 was carried out as previously described⁸⁵. In brief, 3-day-old seedlings were used for transformation. Transgenic composite plants were transferred to vermiculite in a growth chamber (16 h light/8 h dark, 25 °C, 50% relative humidity). For iron treatments, 7-day-old composite plants were transferred to vermiculite moistened with BD–LN nutrient solution with 10 µM Fe citrate or 200 µM AA-DP/2,2′-bipyridine for 3 d, and then inoculated with *B. diazoefficiens* USDA110, and hairy-root samples were collected for further analysis. Transgenic hairy roots transformed with *UBI_{pro}:BTSa-4×MYC*, *UBI_{pro}:NSP1a-4×MYC*, *NSP1a_{pro}:NSP1a^{WT}-4×MYC* or *NSP1a_{pro}:NSP1a^{SKR}-4×MYC* (with GFP in another cassette) were screened using a portable fluorescence lamp (LUYOR-3415, LUYOR) to visualize GFP fluorescence.

For rhizobium-inoculation treatments, 10-day-old composite plants were inoculated with 30 ml *B. diazoefficiens* USDA110–GFP (OD₆₀₀ = 0.08) and hairy roots were collected at 1 DAI for further analysis. For nodulation assays, 7-day-old hairy roots transformed with *UBI_{pro}:NSP1a-4×MYC* were screened by GFP fluorescence, with one positive transgenic hairy root retained per plant, and plants were grown for 3 d before inoculation. Distilled water and BD–LN nutrient solution were irrigated alternately during the growth of transgenic composite plants. Nodule number was scored at 28 DAI.

Histochemical analysis

Roots or nodules from *BTSa_{pro}:GUS*-expressing composite plants were used for GUS assay. Samples were collected at the specified time points, incubated in X-Gluc solution [50 mM NaH₂PO₄, 50 mM Na₂HPO₄, 0.5 mM K₄(Fe(CN)₆), 0.5 mM K₃(Fe(CN)₆), 1 mg ml⁻¹ X-Gluc and 0.1% (v/v) Triton X-100] under vacuum for 10 min, incubated in staining buffer at 37 °C for 4 h for roots and 7 h for nodules, and then washed three times with 75% (v/v) ethanol.

Observation of infection threads and nodule primordia

To observe infection threads and nodule primordia, sterilized seeds were germinated in vermiculite. Ten-day-old seedlings were inoculated with GUS-tagged *B. diazoefficiens* USDA110. Roots were harvested at 3 and 6 DAI and samples were incubated in X-Gluc solution [50 mM NaH₂PO₄, 50 mM Na₂HPO₄, 0.5 mM K₄(Fe(CN)₆), 0.5 mM K₃(Fe(CN)₆), 1 mg ml⁻¹ X-Gluc and 0.1% (v/v) Triton X-100] under vacuum for 10 min, incubated in staining buffer overnight at 37 °C, and then washed three times with 75% (v/v) ethanol. For quantification of infection threads and nodule primordia, all lateral roots on the taproot 5 cm below the junction of the root and stem were selected to count the number of infection threads and nodule primordia.

In vivo stability of NSP1a

For analysis of the NSP1a protein response to iron in hairy roots, NSP1a–4×MYC was expressed in hairy roots and 7-day-old hairy roots were treated with 200 µM AA-DP or 10 µM Fe citrate for 3 d and then inoculated with *B. diazoefficiens* USDA110. Positive transgenic roots were screened using a portable fluorescence lamp to visualize GFP fluorescence at 1 DAI. Total protein was extracted in IB buffer (50 mM HEPES pH 7.5, 150 mM NaCl, 10 mM EDTA pH 8.0, 1% (v/v) Triton X-100, 10% (w/v) sucrose, 2 mM dithiothreitol, 100 mM PMSF, 1% (w/v) polyvinylpyrrolidone, 1× protease inhibitor cocktail (Sigma) and phosphatase inhibitor cocktail (Sigma)). Ten micrograms of total proteins was subjected to SDS–PAGE and NSP1a–4×MYC was detected with anti-MYC (Sigma, C3956, 1:3,000) antibody, and an anti-Actin (ABclonal, CA009, 1:3,000) antibody was used as a loading control.

For the native abundance of NSP1a, WT and *nsp1a-1* or *btsa btsb* seedlings were grown under NFe or LFe condition as described previously, the root sample was collected at 7 DAI and total protein was extracted in IB buffer. Ten micrograms of total proteins was subjected

to SDS–PAGE and native NSP1a protein was detected with anti-NSP1 antibody (1:1,000), with anti-Actin antibody as a loading control.

The anti-NSP1 antibody was generated against the NSP1a and NSP1b-specific peptides (amino acids 4–19). Rabbit antiserum was generated by ABclonal.

In vivo stability of BTSa in the presence of iron

Protein was transiently expressed in *N. benthamiana* leaves from the *UBI_{pro}:BTSa-3×HA* construct. Transformed leaves were infiltrated with 10 mM MgCl₂ containing 50 µM cycloheximide (CHX) and increasing iron concentrations (0, 50 µM, 100 µM, 200 µM, 400 µM ammonium ferric citrate) or 40 µM deferiprone for 48 h after the initial transformation. Twelve hours after treatment, leaves were collected and total proteins extracted (50 mM Tris–HCl pH 7.5, 150 mM NaCl, 10 mM MgCl₂, 1 mM EDTA, 10 mM NaF, 2 mM Na₃VO₄, 25 mM β-glycerophosphate, 10% (w/v) glycerol, 2 mM dithiothreitol, 1 mM phenylmethylsulfonyl fluoride and 1× protease inhibitor cocktail). Ten micrograms of total proteins was subjected to SDS–PAGE and BTSa–3×HA was detected with anti-HA (Sigma, SAB4300603, 1:2,500) antibody, with anti-Actin (ABclonal, CA009, 1:3,000) antibody as a loading control.

For analysis of BTSa in hairy roots, protein was expressed in hairy roots carrying *UBI_{pro}:BTSa-4×MYC* vector, and then 7-day-old hairy roots were treated with 200 µM AA-DP or 10 µM Fe citrate for 3 d. Samples were screened using a portable fluorescence lamp to visualize GFP fluorescence and total protein was extracted in IB buffer. Ten micrograms of total proteins was subjected to SDS–PAGE and BTSa–4×MYC was detected with anti-MYC (Sigma, C3956, 1:3,000) antibody, with anti-Actin (ABclonal, CA009, 1:3,000) antibody as a loading control.

Cell-free degradation assay

For the NSP1a–GST cell-free protein-degradation assay in extracts from *N. benthamiana* leaves, NSP1a–GST was expressed in and purified from *E. coli* with glutathione resin (GenScript, C42062205A) according to manufacturer guidelines. BTSa–3×HA protein and control were extracted from *N. benthamiana* leaves transiently transformed with *UBI_{pro}:BTSa-3×HA* or *UBI_{pro}:3×HA* (50 mM Tris–HCl pH 7.5, 150 mM NaCl, 10 mM MgCl₂, 1 mM EDTA, 10 mM NaF, 2 mM Na₃VO₄, 25 mM β-glycerophosphate, 10% (w/v) glycerol, 2 mM dithiothreitol, 1 mM phenylmethylsulfonyl fluoride and 1× protease inhibitor cocktail). Thirty micrograms of total *N. benthamiana* protein and 9 µg of recombinant GST–NSP1a were mixed in reaction buffer (20 mM Tris–HCl pH 7.5, 10 mM MgCl₂, 1 mM dithiothreitol and 1 mM ATP), then incubated at 28 °C and sampled at the indicated intervals for western blotting. NSP1a–GST was detected with anti-GST (ABclonal, AE001, 1:3,000) antibody, BTSa–3×HA protein was detected with anti-HA (Sigma, SAB4300603, 1:2,500) antibody, and anti-Actin (ABclonal, CA009, 1:3,000) antibody was used as a loading control. All protein immunodetection experiments were performed independently three times with similar results.

In vitro iron-binding assay

The iron-binding assay was performed as previously described²⁹. Briefly, the coding region of full-length *BTSa* and its truncated variant containing three HHE domains were amplified and cloned into pMal-c2X (MBP) vector to generate a construct for expressing MBP–BTSa. The resulting plasmids, as well as the MBP empty vector as a negative control were introduced into *E. coli* Rosetta. Cultures were grown overnight at 16 °C following incubation with 50 µM isopropyl β-D-1-thiogalactopyranoside. Recombinant proteins were purified using amylose resin (Biolabs, E8021V) according to manufacturer instructions, and EDTA was omitted from the column buffer (200 mM NaCl, 20 mM Tris–HCl, 1 mM dithiothreitol). Purified proteins were quantified using a Bio-Rad Protein Assay kit (Hercules) and purified proteins (0.4–1.0 mg) were subjected to inductively coupled plasma–mass spectrometry analysis for determination of iron content.

ChIP–qPCR assay

For ChIP–qPCR assay, 10-day-old transgenic roots expressing *UBI_{pro}:NSP1a-4×MYC*, *UBI_{pro}:NSP1a^{SKR}-4×MYC* or *UBI_{pro}:4×MYC* were inoculated with *B. diazoefficiens* USDA110 in the presence of 10 μM Fe citrate or 200 μM AA-DP. Transgenic hairy roots were collected at 1 DAI and followed by ChIP–qPCR assay as previously described³². Briefly, 1 g of transgenic hairy roots were cross-linked with 1% (v/v) formaldehyde for 10 min under vacuum, and the reaction was stopped by adding 0.125 M glycine. Next, the hairy roots were ground into fine powder in liquid nitrogen, and their nuclei were isolated. Immunoprecipitation was performed using anti-MYC (Sigma, A7470, c-MYC Agarose Affinity Gel) antibody at 4 °C overnight. Immunoprecipitation without the use of antibodies was used as a negative control, while chromatin isolated before precipitation was used as an input control. Soybean *ELF1b* was used as an internal control. Primers used in this experiment are listed in Supplementary Table 2.

In vitro ubiquitination assay

Ubiquitination assays were performed as previously described⁷⁰. Reactions containing 100 nM UBE1 (Boston Biochem, E-305), 2.5 μM UBE2D2 (Boston Biochem, E2-622), 1 μg MBP–BTSa–C^{Ter} and 1 μg 3×HA–ubiquitin (Boston Biochem, U-110) in the ubiquitination reaction buffer [50 mM Tris pH 7.5, 2 mM ATP (Sigma), 5 mM MgCl₂, 2 mM dithiothreitol] were incubated at 30 °C for 3 h. Ubiquitinated proteins were detected by western blotting with anti-HA (Sigma, SAB4300603, 1:2,500) and anti-MBP (ABclonal, AE016, 1:3,000) antibodies.

In vivo ubiquitination assay

In vivo ubiquitination assays were performed as previously described with modifications⁷⁰. MYC-tagged UBQ (UBQ–6×MYC) or a vector control were co-transfected with FLAG-tagged NSP1a (NSP1a–3×FLAG) and HA-tagged BTSa (BTSa–3×HA) into *N. benthamiana* leaves by infiltration of *A. rhizogenes* strain K599. After 2 d of transient expression, *N. benthamiana* leaves were collected for Co-IP assay. Total proteins were extracted with Co-IP buffer [50 mM Tris–HCl pH 7.5, 150 mM NaCl, 5 mM EDTA, 0.5% (v/v) Triton, 1× protease inhibitor cocktail, and before use, adding 1 μM dithiothreitol, 2 μM NaF and 2 μM Na₃VO₃] and incubated with anti-MYC beads (A7470, c-MYC Agarose Affinity Gel, Sigma) at 4 °C for 3 h. Beads were washed five times with Co-IP buffer and the immunoprecipitated proteins were eluted with SDS loading buffer with boiling for 5 min. Samples were analysed by western blotting with anti-FLAG (Sigma, A8592, 1:2,500), anti-MYC (Sigma, C3956, 1:2,000) and anti-HA (Sigma, SAB4300603, 1:2,500) antibodies.

In vitro deubiquitination assay

Mouse USP2-cc was cloned by PCR amplification from mouse cDNA (primers listed in Supplementary Table 2), followed by the method above to construct GST–USP2-cc. GST–USP2-cc fusion proteins were expressed in *E. coli* Rosetta and purified with glutathione resin (Genscript, C42062205A) according to manufacturer guidelines. Deubiquitination assays were performed as previously described with modifications^{70,71}. Briefly, an in vivo ubiquitination assay was performed as described above. After the immunoprecipitation, precipitated complexes were washed once with 50 mM Tris–HCl pH 7.5 and once with DUB dilution buffer (25 mM Tris–HCl pH 7.5, 150 mM NaCl and 10 mM dithiothreitol), aliquoted into individual tubes containing USP2-cc or heat-inactivated (HI) (95 °C for 5 min) USP2-cc as a control in the DUB reaction buffer (50 mM Tris–HCl pH 7.5, 50 mM NaCl and 5 mM dithiothreitol), and incubated at 28 °C overnight. Samples were denatured in SDS buffer and analysed by immunoblotting.

Identification of NSP1a ubiquitination sites in vivo

To identify the ubiquitination sites of NSP1a in vivo, 10-day-old transgenic roots expressing *UBI_{pro}:NSP1a-4×MYC* were inoculated with *B. diazoefficiens* USDA110. Roots were harvested at 1 DAI and 4 g of tissue

was ground to a fine powder in liquid nitrogen. Proteins were extracted by adding 4 ml of extraction buffer (50 mM HEPES pH 7.5, 150 mM NaCl, 10 mM EDTA pH 8.0, 1% (v/v) Triton X-100, 10% (w/v) sucrose, 2 mM dithiothreitol, 100 mM PMSF, 1% (w/v) polyvinylpyrrolidone, protease inhibitor cocktail (Sigma) and phosphatase inhibitor cocktail 2 (Sigma)). Samples were gently mixed for 2 h at 4 °C and centrifuged at 12,000 g for 20 min at 4 °C. Pre-equilibrated anti-MYC beads (Sigma, A7470, c-MYC Agarose Affinity Gel) were then added to the supernatant and the sample was gently agitated at 4 °C for 4 h. The beads were washed five times with extraction buffer, resuspended in 1× SDS loading buffer and separated by SDS–PAGE. Gels were stained with Coomassie blue and the target band was cut for mass spectrometry analysis.

DNA–protein pull-down assay

DNA-binding assays were performed as previously described³⁶. Briefly, biotinylated DNA fragments corresponding to the *NIN1a* promoter were generated by PCR. For the DNA–protein pull-down assay, the biotinylated promoter was incubated with Streptavidin Mag Beads (GenScript, L00424) in TES buffer (10 mM Tris–HCl pH 7.5, 1 mM EDTA pH 8.0, 2 M NaCl) for 1 h at room temperature and washed three times in IP100 buffer (50 mM Tris–HCl pH 7.6, 100 mM potassium glutamate, 2 mM MgCl₂, 0.05% (v/v) Nonidet P40), and then blocked with 5% (w/v) bovine serum albumin for 1 h at room temperature. GST–NSP1a, GST–NSP1a^{SKR} or GST–EV were incubated with DNA-bound beads and rotated at 4 °C for 2 h. Beads were washed three times with IP100 buffer, and then proteins were removed from the beads by boiling in 1× SDS loading buffer and subjected to PAGE. NSP1a–GST was detected with anti-GST (ABclonal, AE001, 1:3,000) antibody. Pull-down efficiency was quantified as the ratio of the NSP1a–GST band in the pull down to the input, and the relative pull-down efficiency represents the normalized efficiency relative to NSP1a^{WT}.

Statistical analysis and reproducibility

Data were analysed using GraphPad Prism 9. Statistical significance was determined using analysis of variance (ANOVA) with Tukey's post hoc test or a two-sided Student's *t*-test, and the statistical test used is described in the figure legends. The experiments were performed three times independently with similar results.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this study are available in the Article and its Supplementary Information files. Source data are provided with this paper.

References

1. Taylor, B. N. & Menge, D. N. L. Light regulates tropical symbiotic nitrogen fixation more strongly than soil nitrogen. *Nat. Plants* **4**, 655–661 (2018).
2. Zheng, M., Zhou, Z., Luo, Y., Zhao, P. & Mo, J. Global pattern and controls of biological nitrogen fixation under nutrient enrichment: a meta-analysis. *Glob. Change Biol.* **25**, 3018–3030 (2019).
3. Elser, J. J. et al. Global analysis of nitrogen and phosphorus limitation of primary producers in freshwater, marine and terrestrial ecosystems. *Ecol. Lett.* **10**, 1135–1142 (2007).
4. LeBauer, D. S. & Treseder, K. K. Nitrogen limitation of net primary productivity in terrestrial ecosystems is globally distributed. *Ecology* **89**, 371–379 (2008).
5. Dénarié, J. & Cullimore, J. Lipo-oligosaccharide nodulation factors: a new class of signaling molecules mediating recognition and morphogenesis. *Cell* **74**, 951–954 (1993).
6. Radutoiu, S. et al. Plant recognition of symbiotic bacteria requires two LysM receptor-like kinases. *Nature* **425**, 585–592 (2003).

7. Limpens, E. et al. LysM domain receptor kinases regulating rhizobial Nod factor-induced infection. *Science* **302**, 630–633 (2003).
8. Madsen, E. B. et al. A receptor kinase gene of the LysM type is involved in legume perception of rhizobial signals. *Nature* **425**, 637–640 (2003).
9. Arrighi, J. F. et al. The *Medicago truncatula* lysin [corrected] motif-receptor-like kinase gene family includes NFP and new nodule-expressed genes. *Plant Physiol.* **142**, 265–279 (2006).
10. Cullimore, J. & Dénarié, J. Plant sciences. How legumes select their sweet talking symbionts. *Science* **302**, 575–578 (2003).
11. Smit, P. et al. NSP1 of the GRAS protein family is essential for rhizobial Nod factor-induced transcription. *Science* **308**, 1789–1791 (2005).
12. Kaló, P. et al. Nodulation signaling in legumes requires NSP2, a member of the GRAS family of transcriptional regulators. *Science* **308**, 1786–1789 (2005).
13. Schauser, L., Roussis, A., Stiller, J. & Stougaard, J. A plant regulator controlling development of symbiotic root nodules. *Nature* **402**, 191–195 (1999).
14. Hirsch, S. et al. GRAS proteins form a DNA binding complex to induce gene expression during nodulation signaling in *Medicago truncatula*. *Plant Cell* **21**, 545–557 (2009).
15. Ané, J. M. et al. *Medicago truncatula* DMI1 required for bacterial and fungal symbioses in legumes. *Science* **303**, 1364–1367 (2004).
16. Imaizumi-Anraku, H. et al. Plastid proteins crucial for symbiotic fungal and bacterial entry into plant roots. *Nature* **433**, 527–531 (2005).
17. Charpentier, M. et al. *Lotus japonicus* CASTOR and POLLUX are ion channels essential for perinuclear calcium spiking in legume root endosymbiosis. *Plant Cell* **20**, 3467–3479 (2008).
18. Charpentier, M. et al. Nuclear-localized cyclic nucleotide-gated channels mediate symbiotic calcium oscillations. *Science* **352**, 1102–1105 (2016).
19. Capoen, W. et al. Nuclear membranes control symbiotic calcium signaling of legumes. *Proc. Natl Acad. Sci. USA* **108**, 14348–14353 (2011).
20. Desbrosses, G. J. & Stougaard, J. Root nodulation: a paradigm for how plant-microbe symbiosis influences host developmental pathways. *Cell Host Microbe* **10**, 348–358 (2011).
21. Downie, J. A. Legume nodulation. *Curr. Biol.* **24**, R184–R190 (2014).
22. Caetano-Anollés, G. & Gresshoff, P. M. Plant genetic control of nodulation. *Annu. Rev. Microbiol.* **45**, 345–382 (1991).
23. Ferguson, B. J. et al. Molecular analysis of legume nodule development and autoregulation. *J. Integr. Plant Biol.* **52**, 61–76 (2010).
24. Ferguson, B. J. et al. Legume nodulation: the host controls the party. *Plant Cell Environ.* **42**, 41–51 (2019).
25. Soyano, T., Hirakawa, H., Sato, S., Hayashi, M. & Kawaguchi, M. Nodule inception creates a long-distance negative feedback loop involved in homeostatic regulation of nodule organ production. *Proc. Natl Acad. Sci. USA* **111**, 14607–14612 (2014).
26. Tsikou, D. et al. Systemic control of legume susceptibility to rhizobial infection by a mobile microRNA. *Science* **362**, 233–236 (2018).
27. Searle, I. R. et al. Long-distance signaling in nodulation directed by a CLAVATA1-like receptor kinase. *Science* **299**, 109–112 (2003).
28. Wang, L. et al. A GmNINa-miR172c-NNC1 regulatory network coordinates the nodulation and autoregulation of nodulation pathways in soybean. *Mol. Plant* **12**, 1211–1226 (2019).
29. Takahara, M. et al. Too much love, a novel Kelch repeat-containing F-box protein, functions in the long-distance regulation of the legume–*Rhizobium* symbiosis. *Plant Cell Physiol.* **54**, 433–447 (2013).
30. Malik, N. S., Calvert, H. E. & Bauer, W. D. Nitrate induced regulation of nodule formation in soybean. *Plant Physiol.* **84**, 266–271 (1987).
31. Wang, T. et al. Light-induced mobile factors from shoots regulate rhizobium-triggered soybean root nodulation. *Science* **374**, 65–71 (2021).
32. Ji, H. et al. Differential light-dependent regulation of soybean nodulation by papilionoid-specific HY5 homologs. *Curr. Biol.* **32**, 783–795.e5 (2022).
33. Li, X. et al. Shoot-to-root translocated GmNIN1/FT2a triggers nodulation and regulates soybean nitrogen nutrition. *PLoS Biol.* **20**, e3001739 (2022).
34. Lin, J. et al. Zinc mediates control of nitrogen fixation via transcription factor filamentation. *Nature* **631**, 164–169 (2024).
35. González-Guerrero, M. et al. Forging a symbiosis: transition metal delivery in symbiotic nitrogen fixation. *New Phytol.* **239**, 2113–2125 (2023).
36. Wu, X. et al. GmYSL7 controls iron uptake, allocation, and cellular response of nodules in soybean. *J. Integr. Plant Biol.* **65**, 167–187 (2023).
37. Siegl, A., Afjehi-Sadat, L. & Wienkoop, S. Systemic long-distance sulfur transport and its role in symbiotic root nodule protein turnover. *J. Plant Physiol.* **297**, 154260 (2024).
38. Liang, G. Iron uptake, signaling, and sensing in plants. *Plant Commun.* **3**, 100349 (2022).
39. Hänsch, R. & Mendel, R. R. Physiological functions of mineral micronutrients (Cu, Zn, Mn, Fe, Ni, Mo, B, Cl). *Curr. Opin. Plant Biol.* **12**, 259–266 (2009).
40. Gallon, J. R. The oxygen sensitivity of nitrogenase: a problem for biochemists and micro-organisms. *Trends Biochem. Sci.* **6**, 19–23 (1981).
41. Peters, J. W. & Szilagyi, R. K. Exploring new frontiers of nitrogenase structure and mechanism. *Curr. Opin. Chem. Biol.* **10**, 101–108 (2006).
42. Tang, C., Robson, A. D. & Dilworth, M. J. The role of iron in nodulation and nitrogen fixation in *Lupinus angustifolius* L. *New Phytol.* **114**, 173–182 (1990).
43. Tang, C., Robson, A. D. & Dilworth, M. J. Which stage of nodule initiation in *Lupinus angustifolius* L. is sensitive to iron deficiency? *New Phytol.* **117**, 243–250 (1991).
44. Tang, C., Robson, A. D. & Dilworth, M. J. A split-root experiment shows that iron is required for nodule initiation in *Lupinus angustifolius* L. *New Phytol.* **115**, 61–67 (1990).
45. Brear, E. M., Day, D. A. & Smith, P. M. Iron: an essential micronutrient for the legume–rhizobium symbiosis. *Front. Plant Sci.* **4**, 359 (2013).
46. Tabata, R. Regulation of the iron-deficiency response by IMA/FEP peptide. *Front. Plant Sci.* **14**, 1107405 (2023).
47. Jakoby, M., Wang, H. Y., Reidt, W., Weisshaar, B. & Bauer, P. FRU (BHLH029) is required for induction of iron mobilization genes in *Arabidopsis thaliana*. *FEBS Lett.* **577**, 528–534 (2004).
48. Colangelo, E. P. & Guerinot, M. L. The essential basic helix-loop-helix protein FIT1 is required for the iron deficiency response. *Plant Cell* **16**, 3400–3412 (2004).
49. Yuan, Y. et al. FIT interacts with AtbHLH38 and AtbHLH39 in regulating iron uptake gene expression for iron homeostasis in *Arabidopsis*. *Cell Res.* **18**, 385–397 (2008).
50. Wang, N. et al. Requirement and functional redundancy of lb subgroup bHLH proteins for iron deficiency responses and uptake in *Arabidopsis thaliana*. *Mol. Plant* **6**, 503–513 (2013).
51. Liang, G., Zhang, H., Li, X., Ai, Q. & Yu, D. bHLH transcription factor bHLH115 regulates iron homeostasis in *Arabidopsis thaliana*. *J. Exp. Bot.* **68**, 1743–1755 (2017).
52. Zhang, J. et al. The bHLH transcription factor bHLH104 interacts with IAA-LEUCINE RESISTANT3 and modulates iron homeostasis in *Arabidopsis*. *Plant Cell* **27**, 787–805 (2015).

53. Gao, F. et al. The transcription factor bHLH121 interacts with bHLH105 (ILR3) and its closest homologs to regulate iron homeostasis in *Arabidopsis*. *Plant Cell* **32**, 508–524 (2020).
54. Kim, S. A., LaCroix, I. S., Gerber, S. A. & Guerinot, M. L. The iron deficiency response in *Arabidopsis thaliana* requires the phosphorylated transcription factor URI. *Proc. Natl Acad. Sci. USA* **116**, 24933–24942 (2019).
55. Hindt, M. N. et al. BRUTUS and its paralogs, BTS LIKE1 and BTS LIKE2, encode important negative regulators of the iron deficiency response in *Arabidopsis thaliana*. *Metallomics* **9**, 876–890 (2017).
56. Kobayashi, T. et al. Iron-binding haemerythrin RING ubiquitin ligases regulate plant iron responses and accumulation. *Nat. Commun.* **4**, 2792 (2013).
57. Selote, D., Samira, R., Matthiadis, A., Gillikin, J. W. & Long, T. A. Iron-binding E3 ligase mediates iron response in plants by targeting basic helix-loop-helix transcription factors. *Plant Physiol.* **167**, 273–286 (2015).
58. Rodríguez-Celma, J. et al. *Arabidopsis* BRUTUS-LIKE E3 ligases negatively regulate iron uptake by targeting transcription factor FIT for recycling. *Proc. Natl Acad. Sci. USA* **116**, 17584–17591 (2019).
59. Li, Y. et al. IRON MAN interacts with BRUTUS to maintain iron homeostasis in *Arabidopsis*. *Proc. Natl Acad. Sci. USA* **118**, e2109063118 (2021).
60. Riaz, N. & Guerinot, M. L. All together now: regulation of the iron deficiency response. *J. Exp. Bot.* **72**, 2045–2055 (2021).
61. Cao, M. et al. Spatial IMA1 regulation restricts root iron acquisition on MAMP perception. *Nature* **625**, 750–759 (2024).
62. Ganz, T. & Nemeth, E. Iron homeostasis in host defence and inflammation. *Nat. Rev. Immunol.* **15**, 500–510 (2015).
63. Ito, M. et al. IMA peptides regulate root nodulation and nitrogen homeostasis by providing iron according to internal nitrogen status. *Nat. Commun.* **15**, 733 (2024).
64. Wu, Z. et al. A global coexpression network of soybean genes gives insights into the evolution of nodulation in nonlegumes and legumes. *New Phytol.* **223**, 2104–2119 (2019).
65. Shimomura, K., Nomura, M., Tajima, S. & Kouchi, H. LjnsRING, a novel RING finger protein, is required for symbiotic interactions between *Mesorhizobium loti* and *Lotus japonicus*. *Plant Cell Physiol.* **47**, 1572–1581 (2006).
66. Onaga, G., Dramé, K. N. & Ismail, A. M. Understanding the regulation of iron nutrition: can it contribute to improving iron toxicity tolerance in rice? *Funct. Plant Biol.* **43**, 709–726 (2016).
67. Chen, J. et al. The B-type response regulator GmRR11d mediates systemic inhibition of symbiotic nodulation. *Nat. Commun.* **13**, 7661 (2022).
68. Hider, R. C. & Hoffbrand, A. V. The role of deferiprone in iron chelation. *N. Engl. J. Med.* **379**, 2140–2150 (2018).
69. Dikic, I. & Schulman, B. A. An expanded lexicon for the ubiquitin code. *Nat. Rev. Mol. Cell Biol.* **24**, 273–287 (2023).
70. Ma, X. et al. Ligand-induced monoubiquitination of BIK1 regulates plant immunity. *Nature* **581**, 199–203 (2020).
71. Hospenhal, M. K., Mevissen, T. E. T. & Komander, D. Deubiquitinase-based analysis of ubiquitin chain architecture using Ubiquitin Chain Restriction (UbiCRest). *Nat. Protoc.* **10**, 349–361 (2015).
72. Huang, Y. et al. Improving rice nitrogen-use efficiency by modulating a novel monoubiquitination machinery for optimal root plasticity response to nitrogen. *Nat. Plants* **9**, 1902–1914 (2023).
73. Pavri, R. et al. Histone H2B monoubiquitination functions cooperatively with FACT to regulate elongation by RNA polymerase II. *Cell* **125**, 703–717 (2006).
74. Slatni, T. et al. Growth, nitrogen fixation and ammonium assimilation in common bean (*Phaseolus vulgaris* L.) subjected to iron deficiency. *Plant Soil* **312**, 49–57 (2008).
75. O'Hara, G. W., Dilworth, M. J., Boonkerd, N. & Parkpian, P. Iron-deficiency specifically limits nodule development in peanut inoculated with *Bradyrhizobium* sp. *New Phytol.* **108**, 51–57 (1988).
76. Li, X. R. et al. Nutrient regulation of lipochitooligosaccharide recognition in plants via NSP1 and NSP2. *Nat. Commun.* **13**, 6421 (2022).
77. Hohnjec, N., Czaja-Hasse, L. F., Hogeckamp, C. & Küster, H. Pre-announcement of symbiotic guests: transcriptional reprogramming by mycorrhizal lipochitooligosaccharides shows a strict co-dependency on the GRAS transcription factors NSP1 and RAM1. *BMC Genomics* **16**, 994 (2015).
78. Delaux, P. M., Bécard, G. & Combier, J. P. NSP1 is a component of the Myc signaling pathway. *New Phytol.* **199**, 59–65 (2013).
79. Liu, J. et al. Salicylic acid involved in chilling-induced accumulation of calycosin-7-O- β -D-glucoside in *Astragalus membranaceus* adventitious roots. *Acta Physiol. Plant.* **41**, 120 (2019).
80. Ma, X. et al. A robust CRISPR/Cas9 system for convenient, high-efficiency multiplex genome editing in monocot and dicot plants. *Mol. Plant* **8**, 1274–1284 (2015).
81. Lei, Y. et al. CRISPR-P: a web tool for synthetic single-guide RNA design of CRISPR-system in plants. *Mol. Plant* **7**, 1494–1496 (2014).
82. Zhang, Z., Xing, A., Staswick, P. & Clemente, T. E. The use of glufosinate as a selective agent in *Agrobacterium*-mediated transformation of soybean. *Plant Cell Tissue Organ Cult.* **56**, 37–46 (1999).
83. Yu, H. et al. GmNAC039 and GmNAC018 activate the expression of cysteine protease genes to promote soybean nodule senescence. *Plant Cell* **35**, 2929–2951 (2023).
84. Porebski, S., Bailey, L. G. & Baum, B. R. Modification of a CTAB DNA extraction protocol for plants containing high polysaccharide and polyphenol components. *Plant Mol. Biol. Rep.* **15**, 8–15 (1997).
85. Kereszt, A. et al. *Agrobacterium rhizogenes*-mediated transformation of soybean to study root biology. *Nat. Protoc.* **2**, 948–952 (2007).

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Author contributions

X.L. and Z.W. designed the experiments. Z.R., L.Z., H.L., M.Y., Xuesong Wu, R.H., J.L., H.W. and Xinying Wu performed the experiments. X.L., Z.W. and Z.R. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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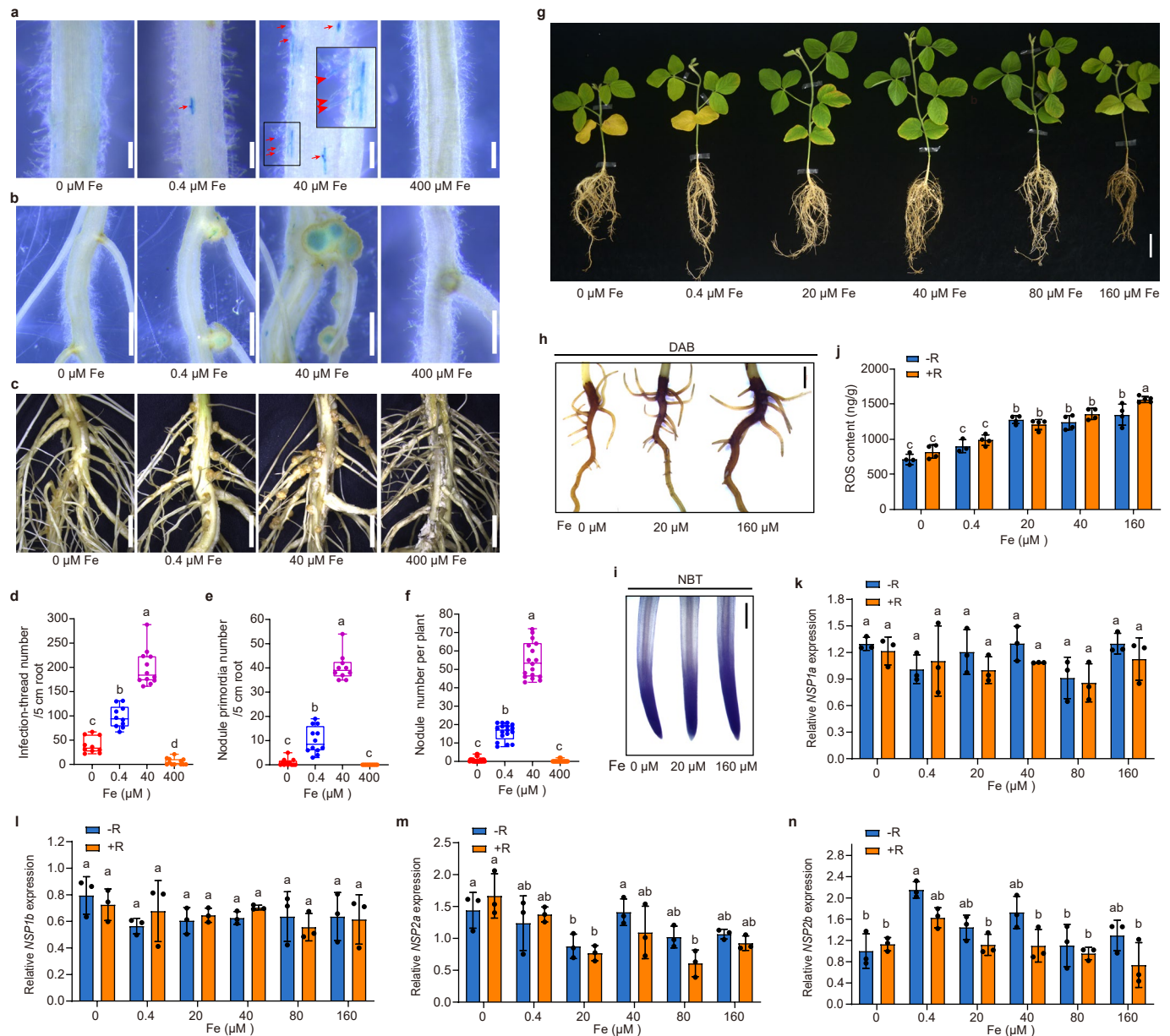
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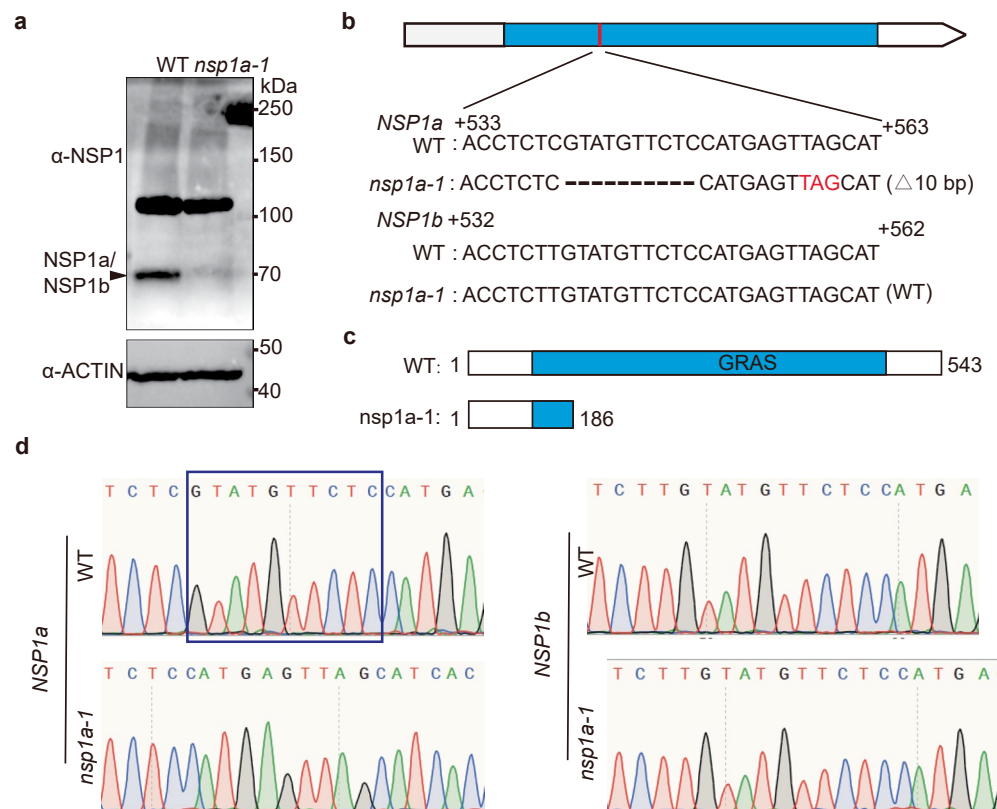
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Extended Data Fig. 1 | Iron levels determine nodulation success in soybean.

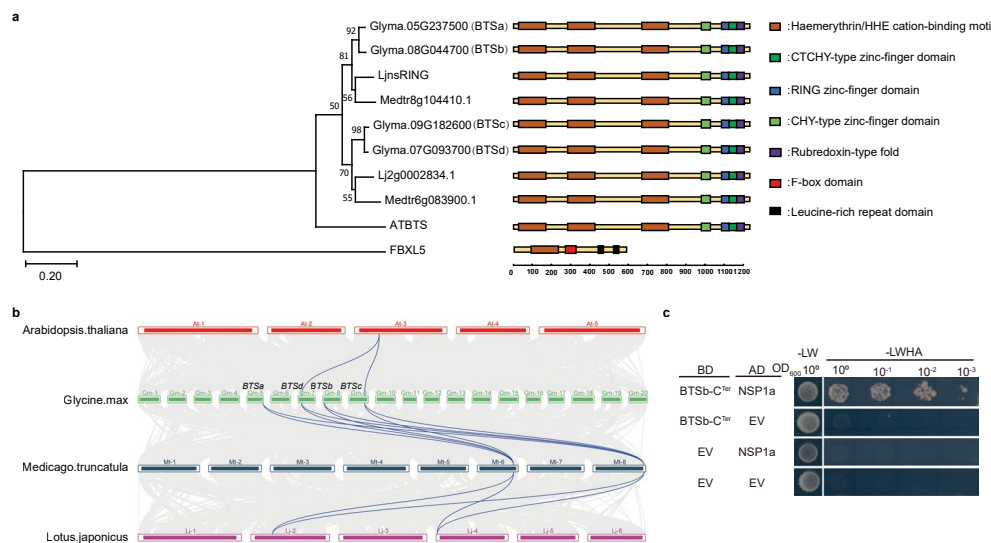
a and **d**, Representative images (a) and number (d) of infection threads in wild-type Wm82 roots at 3 DAI. Scale bars = 200 μ m. Data are the means $\pm$ SD of $n = 10, 10, 12, 12$ for 0, 0.4, 40, 400 μ M Fe. **b** and **e**, Representative images (b) and number (e) of nodule primordia in Wm82 roots at 6 DAI. Scale bars = 1 mm. Data are the means $\pm$ SD of $n = 12, 12, 10, 12$ for 0, 0.4, 40, 400 μ M Fe. **c** and **f**, Representative images of nodules (c) and Nodule numbers (f) at 18 DAI. Scale bars = 5 mm. Data are the means $\pm$ SD of $n = 26, 17, 17, 19$ for 0, 0.4, 40, 400 μ M Fe. **g**, Phenotypes of soybean plants under different iron conditions. Plants were imaged at 21 DAI. Scale bars = 5 cm. **h** and **i**, DAB staining (h) and NBT staining (i) of roots at 1 DAI.

Scale bar = 5 mm in (h) and 0.5 mm in (i). **j**, ROS content in roots at 1 DAI. Data are means $\pm$ SD of $n = 3$ (uninoculated 0.4 μ M), $n = 5$ (inoculated 160 μ M) and $n = 4$ (the others). **k-n**, Expression of *NSP1a* (k), *NSP1b* (l), *NSP2a* (m) and *NSP2b* (n) under different iron conditions at 1 DAI. Data are means $\pm$ SD of $n = 3$. For the box-plot in **d**, **e** and **f**, whiskers indicate minimum and maximum values, line indicates the median, the box boundaries indicate the upper (25th percentile) and lower (75th percentile) quartiles. **d-f**, Significant differences were determined by one-way ANOVA ($P < 0.05$); **j**, **k-n**, significant differences were determined by two-way ANOVA ($P < 0.05$). The experiments were performed for three times with similar results.



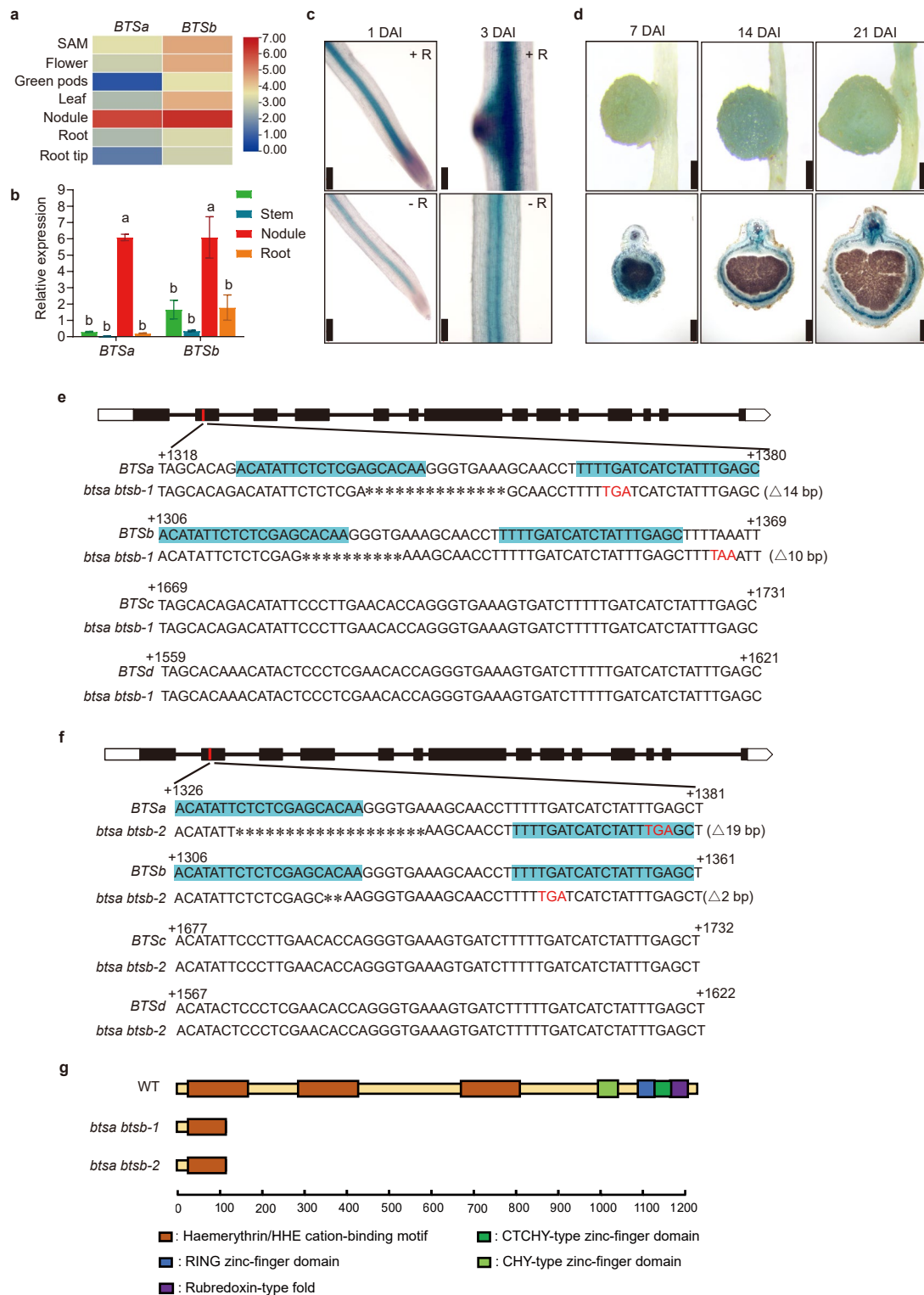
Extended Data Fig. 2 | Detection of native NSP1 by anti-NSP1 antibody and CRISPR-Cas9 gene editing of *NSP1a*. **a**, Western blot of native NSP1a abundance in nodules of WT and *nsp1a-1* mutant using anti-NSP1 antibody. The arrow indicates the specific band of NSP1a and NSP1b. The experiment was repeated for three times with similar results. **b**, Schematic of the *NSP1a* gene. The *nsp1a-1* mutant in the Wm82 background has a 10-bp (540–549 bp) deletion in the

coding region induced by CRISPR-Cas9. The red font in the mutant represents the new translational stop codon due to a frame shift arising from Cas9 nuclease activity. **c**, Diagram of wild-type and mutant NSP1a protein. **d**, Sanger sequencing chromatograms of wild-type Wm82 and *nsp1a-1* mutant. The blue box indicates deleted region in the *nsp1a-1* mutant.



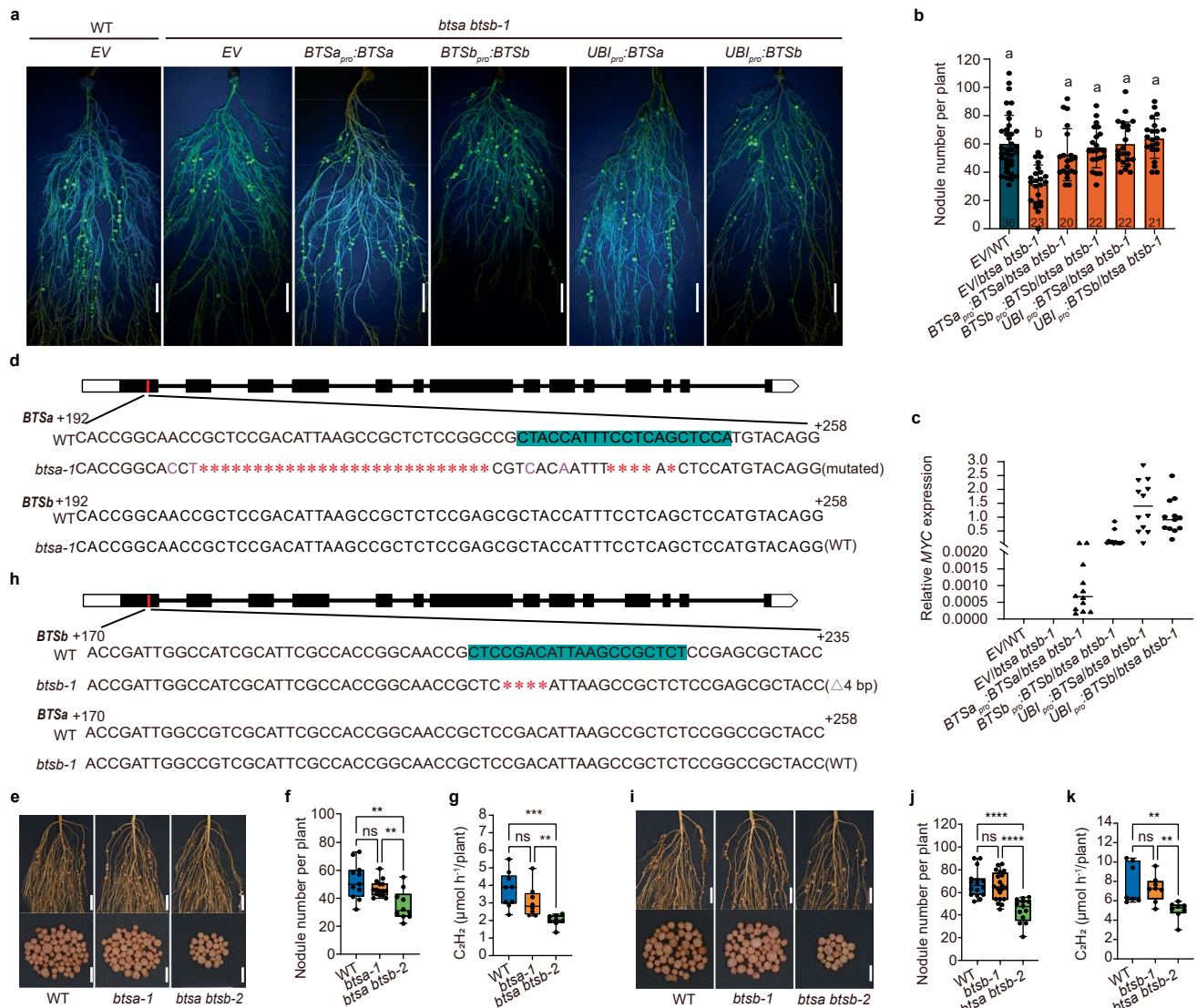
Extended Data Fig. 3 | Phylogenetic analysis of BTS orthologs and interaction assay between BTSb and NSP1a. **a**, Phylogenetic tree and domain structures of select BTS orthologs. Clustal-W was used to align orthologs and homologs from *Glycine max*, *Medicago truncatula*, *Lotus japonicus* and *Arabidopsis thaliana*. MEGA-X was used to construct the phylogenetic tree with 1,000 bootstrap replications. **b**, Reference-free alignment of BTS proteins in *Arabidopsis thali-*

ana (At), *Glycine max* (Gm), *Medicago truncatula* (Mt) and *Lotus japonicus* (Lj) generated by TBtools. **c**, Yeast two-hybrid assay of the BTSb C-terminal domain (BTSb-C^{ter}) and NSP1a. BD: Binding domain. AD: Activation domain. -LW: Leu and Trp dropout. -LWHA: Leu, Trp, His and Ade dropout. The experiment was repeated for three times with similar results.



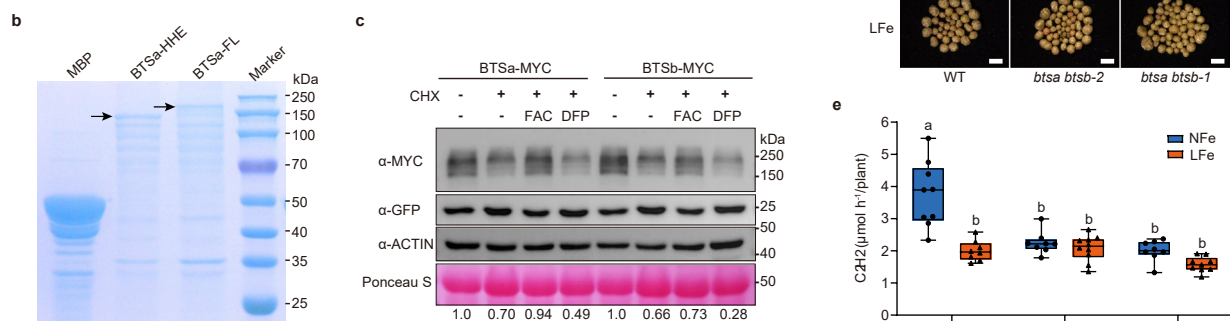
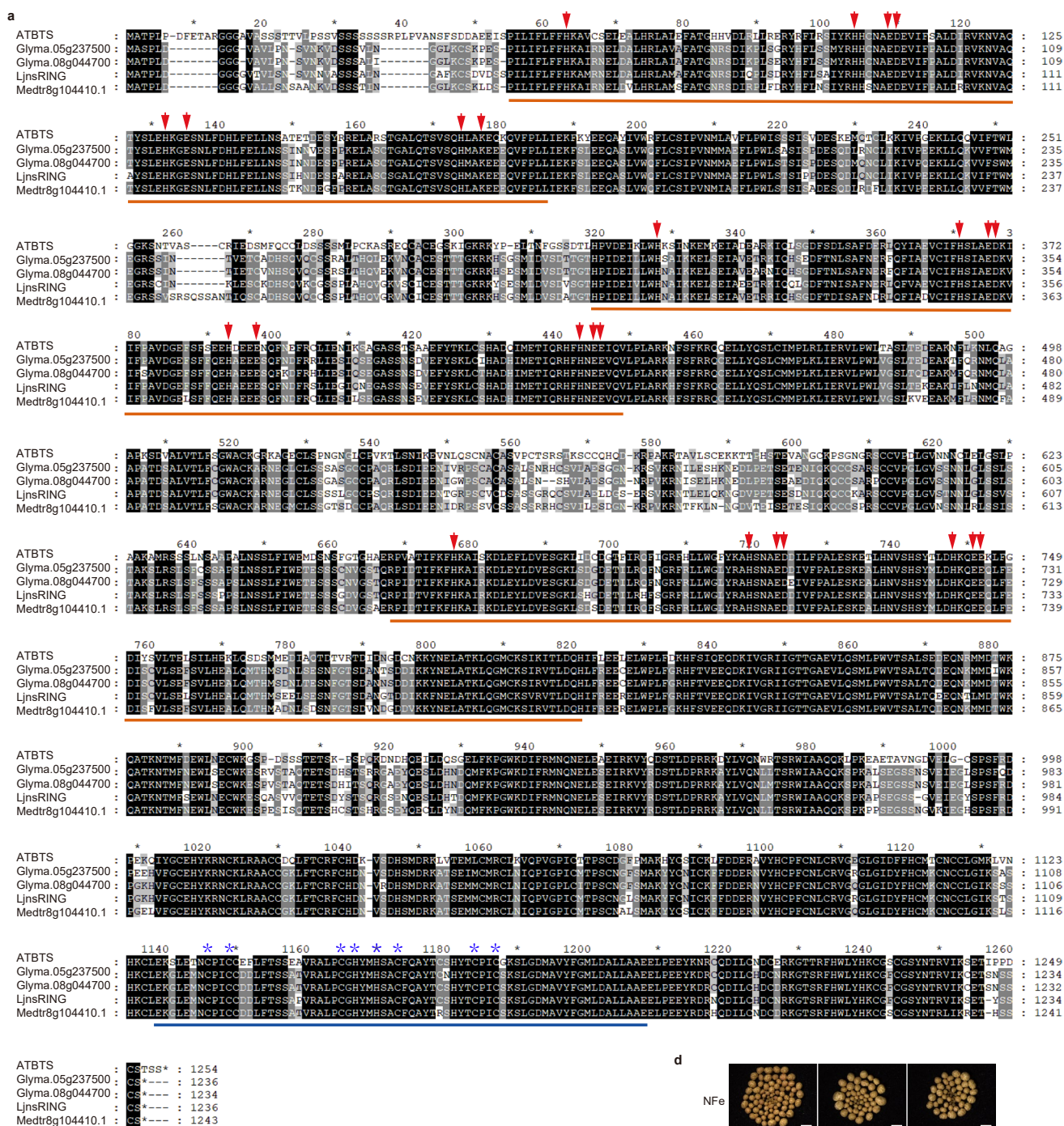
Extended Data Fig. 4 | *BTSa* and *BTSb* expression assay and CRISPR–Cas9 editing of *BTSa* and *BTSb*. **a**, Expression profile of soybean *BTSa* and *BTSb* genes using data deposited on Phytozome. A heatmap was generated by TBtools. The color scale shows increasing expression levels from blue to red. **b**, RT–qPCR analysis of *BTSa* and *BTSb* in different tissues. Samples were collected at 21 DAI. Expression was normalized to *ELF1b* and values are means of three technical repetitions and SEM. The experiment was repeated for three times with similar results. **c** and **d**, Histochemical staining of GUS in the hairy roots with (+ R) or without (- R) inoculation (c) and nodules at different developmental stages (d) expressing *BTSa_{pro}:GUS*. Scale bars = 0.5 mm. The experiment was repeated for

three times with similar results. **e** and **f**, CRISPR–Cas9 was used to mutate *BTSa* and *BTSb* using four sgRNAs targeting both *BTSa* and *BTSb* in exon 1 and 2. Two mutants were recovered (assigned *btsa btsb-1* and *btsa btsb-2*) with edits in exon 2. Characterization of *BTSa* and *BTSb* edits in *btsa btsb-1* (e) and *btsa btsb-2* (f). Top: Schematic of the genomic structure of *BTSa*. Bottom: Alignment of *BTSa*, *BTSb*, *BTSb-1* and *BTSb-2* sequences in Wm82, *btsa btsb-1* and *btsa btsb-2*. The sequence in blue represents the sgRNA, asterisks represent deletions detected in the edited allele, and the red font in mutants represent the stop codon arising from frame shifts induced by editing. **g**, Schematic of proteins encoded by Wm82 and the mutants.



Extended Data Fig. 5 | BTSa and BTSb play a similar role in regulating nodulation. **a** and **b**, Nodulation phenotype (**a**) and nodule number (**b**) of hairy roots expressing *BTSa* or *BTSb* at 21 DAI. Scar bars = 2 cm. The numbers in the column of (**b**) represent the number of samples, data are the means $\pm$ SD and significant differences were determined by one-way ANOVA ($P < 0.05$). **c**, *BTSa*-MYC and *BTSb*-MYC expression in hairy roots in (**a**). Data are the means $\pm$ SD of $n = 12$. **d**, Characterization of *BTSa* editing in *btsa-1* mutant. The red stars indicate deletion base pairs. **e** and **f**, Nodulation phenotypes (**e**) and nodule number (**f**) at 21 DAI. Scar bars = 2 cm for roots and 5 mm for nodules. Data are the means $\pm$ SD of $n = 10, 11, 15$ for WT, *btsa-1*, *btsa btsb-2*. **g**, Nitrogenase activity at 21 DAI. Data are the means $\pm$ SD of $n = 9, 8, 8$ for WT, *btsa-1*, *btsa btsb-2*. **h**, Characterization

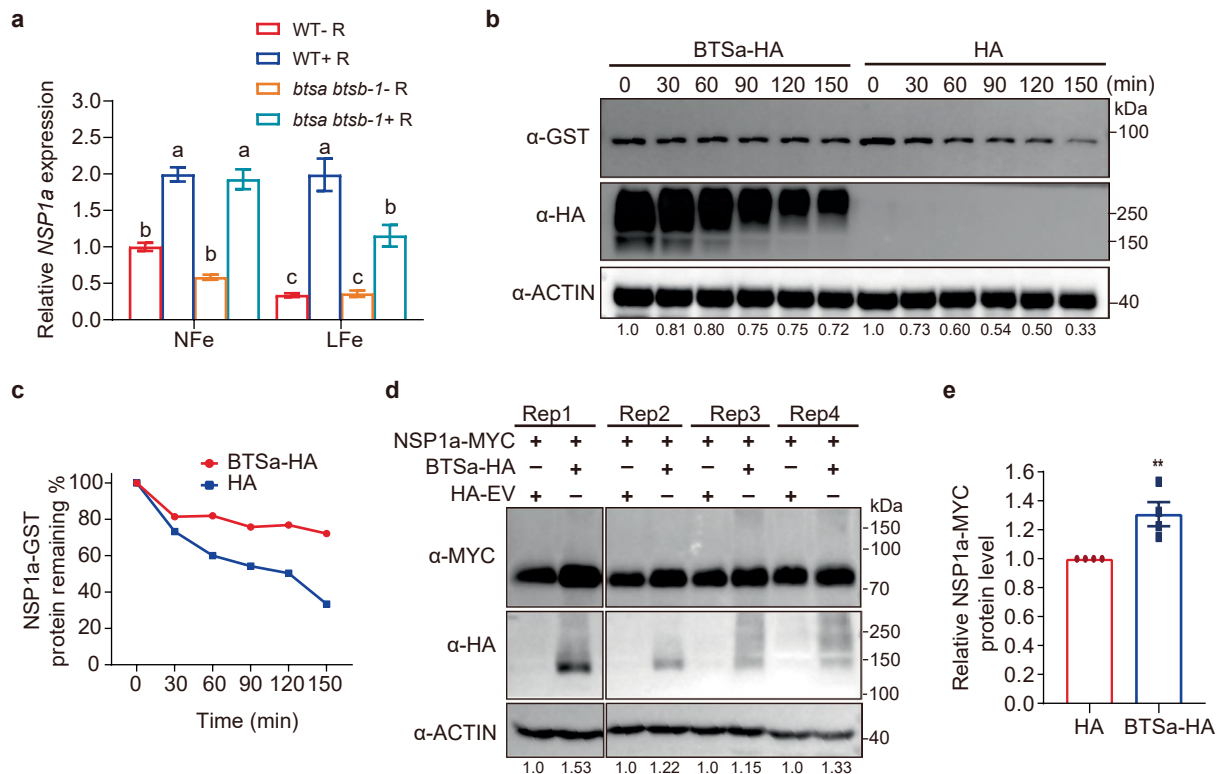
of *BTSb* editing in *btsb-1* mutant. The red stars indicate the deletion base pairs. **i** and **j**, Nodulation phenotypes (**i**) and nodule number (**j**) at 21 DAI. Scar bars = 2 cm for roots and 5 mm for nodules. Data are the means $\pm$ SD of $n = 16, 20, 13$ for WT, *btsa-1*, *btsa btsb-2*. **k**, Nitrogenase activity of WT, *btsa-1* and *btsa btsb-2* at 21 DAI. Data are the means $\pm$ SD of $n = 7, 8, 8$ for WT, *btsa-1*, *btsa btsb-2*. **f**, **g**, **j** and **k**, Significant differences were determined by two-sided Student's *t*-test (** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$). For the box-plot in (**f**, **g**, **j** and **k**), whiskers indicate minimum and maximum values, line indicates the median, the box boundaries indicate the upper (25th percentile) and lower (75th percentile) quartiles. The experiments were performed for three times with similar results.



Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Amino-acid alignment of BTS proteins, detection of BTSa-MBP, stability assay of BTSa-MYC and BTSb-MYC and nodulation phenotypes of *btsa btsb* double mutants. **a**, Amino-acid alignment of the BTS proteins from soybean, *Lotus japonicus*, *Medicago truncatula* and *Arabidopsis thaliana*. Hemerythrin (HHE) domains were marked with an orange underline, conserved histidine, glutamate and aspartate residues that potentially bind to metals are marked with red arrows. The RING zinc-finger domain was marked with a blue underline and the conserved cysteine and histidine are marked with blue asterisks. **b**, Coomassie Blue staining of proteins purified from *E. coli*. The arrows indicate bands of MBP-HHE (with three HHE domains) and MBP-BTSa. **c**, *In vivo* stability of BTSa-MYC and BTSb-MYC under different iron conditions. The BTSa-MYC and BTSb-MYC fusion proteins were expressed in *N. bethamiana* leaf and treated without CHX or with CHX plus 40 μ M FAC or DFP. Anti-GFP was used as control of expression efficiency, anti-Actin or Ponceau S staining

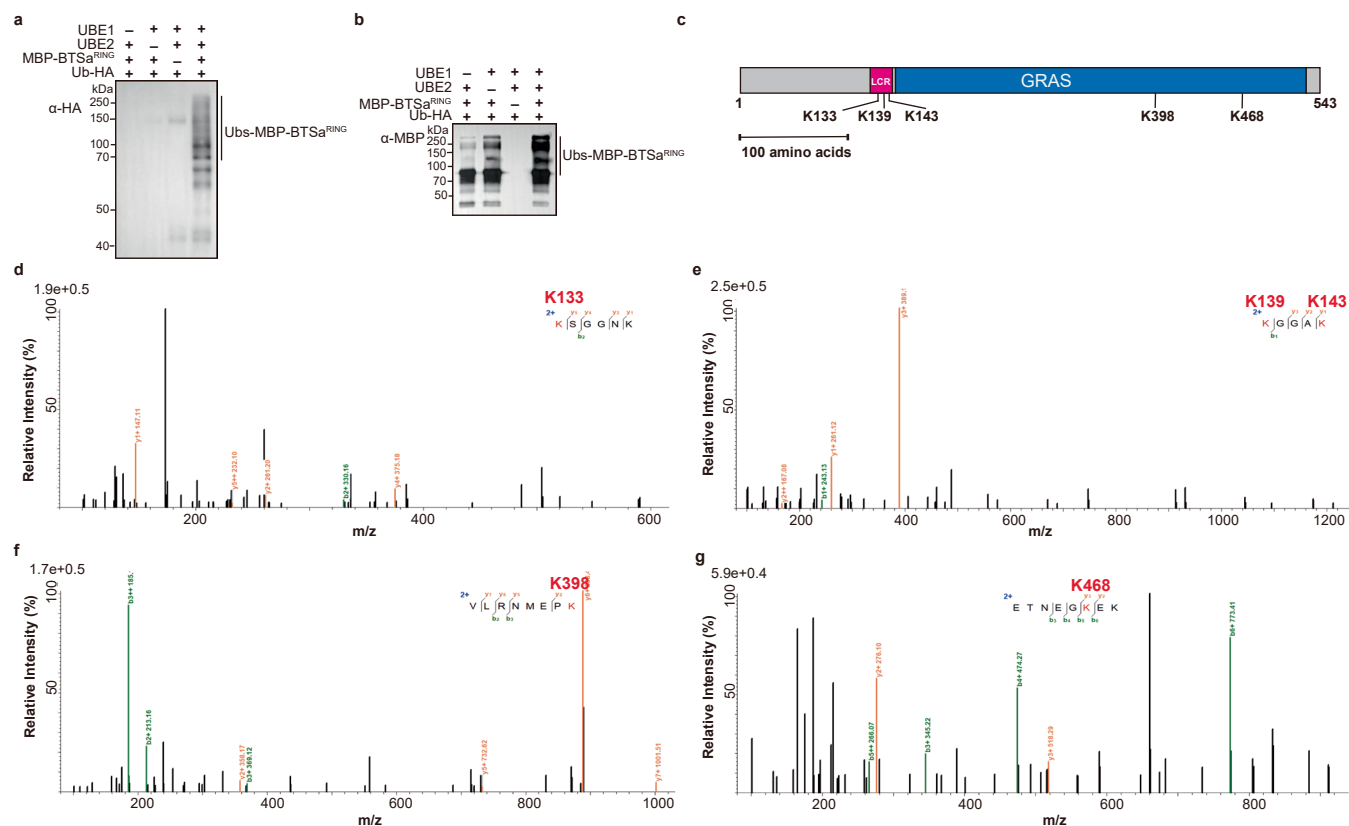
as loading controls. The numbers at the bottom denote the relative intensities normalized to BTSa-MYC intensity in the sample without CHX. FAC: Ferric ammonium citrate, DFP: deferiprone. The experiments were performed for three times with similar results. **d**, Nodulation phenotypes of WT and *btsa btsb* double mutants under different iron conditions. Number of nodules per plant were harvested and evaluated at 21 DAI. Scale bars = 2 mm. **e**, Nitrogenase activity of *btsa btsb* mutants under different iron conditions. Data are the means $\pm$ SD of $n = 7$ (WT in LFe, *btsa btsb-1, 2* in NFe), 9 (WT in NFe, *btsa btsb-2* in LFe) and 10 (*btsa btsb-1* in LFe). Significant differences were determined by two-way ANOVA ($P < 0.05$). For the box-plot, whiskers indicate minimum and maximum values, line indicates the median, the box boundaries indicate the upper (25th percentile) and lower (75th percentile) quartiles. The experiments were performed for three times with similar results.



Extended Data Fig. 7 | Transcriptional and protein stability assays of NSP1a.

a, qRT-PCR analysis of *NSP1a* in uninoculated (-R) and inoculated (+R) roots of WT Wm82 and *btsa btsb-1* grown under NFe and LFe conditions sampled at 1 DAI. Expression was normalized to *ELF1b* and values are means of three technical repetitions and SEM. The experiments were repeated for three times with similar results. Significant differences were determined by two-way ANOVA ($P < 0.05$). NFe: 10 mM Fe-citrate, LFe: 200 mM AA-DP. **b**, Cell-free degradation assay with recombinant GST-NSP1a in protein extracts from *N. bethamiana* leaves transiently transformed with *UBI_{pro}:BTSa-3×HA* or an empty-vector control (HA). The samples were incubated for the indicated times and protein abundance was determined by western blotting. The numbers at the bottom denote the

relative intensities of GST-NSP1a normalized to 0 point. The experiments were repeated for three times with similar results. **c**, Quantification of GST-NSP1a band intensities in (b). **d**, In vivo analysis of NSP1a-MYC stability in the presence or absence of BTSa-HA. *UBI_{pro}:NSP1a-4×MYC* was transiently expressed in *N. bethamiana* leaves with *UBI_{pro}:BTSa-3×HA* or vector control (HA-EV). The samples were collected 48 h after infiltration and prepared for western blotting. The numbers at the bottom denote the relative intensities of NSP1a-MYC normalized to respective HA-EV control. **e**, Quantification of NSP1a-MYC band intensities in (d). Data are the means $\pm$ SD of $n = 4$. Significant difference was determined by two-tailed Student's *t*-test (** $P < 0.01$).



Extended Data Fig. 8 | BTSa possesses E3 ubiquitin-ligase activity assay of BTs and characterization of ubiquitination sites in NSP1a by BTSa. a and b, *In vitro* ubiquitination assay. BTSa^{RING} (E3) fused to MBP was used with (+) or without (-) HA-tagged ubiquitin (3×HA-Ub), human E1 (UBE1) and E2 (UBE2D2). 500 ng MBP-BTSa^{RING} recombinant protein and 500 ng 3×HA-Ubiquitin was loaded onto 10% SDS-PAGE gels and ubiquitinated proteins were detected by

western blotting using an anti-HA antibody in (a) and anti-MBP antibody in (b). The experiment was repeated three times with similar results. **c-g**, Identification of NSP1a ubiquitination sites *in vivo* by LC-MS/MS. MS analysis was performed once. **c**, Schematic of NSP1a and the distribution of its ubiquitination sites. LCR: Low-complexity region. **d-g**, K133, K139, K143, K398 and K468 are potential ubiquitination sites.

a

GmNSP1a:	MTMEPN-PTSDHSIDWFEGSVSFFPFLDYFYDS-GDIQDYQWLEQ--DITSHYQTDNTSSSPNATN-----ISTTTTLETCDSSNNLPLSDLP :	87
GmNSP1b:	MTMEPN-PTPDHSIDWFEGSVSFFPFLDYFYNS-SDIQEYQWLEQ--DIASNYQTDNTSS-PNATN-----IVTITTTLETCDSSNN-LPLSDLP :	85
LjNSP1:	MTMEPNSTTSDHILDWLEGSVSFFPSFLDEPCNNSGYIQEYHLD-----QYQTDANTGSSPNATNSTTATIVATSATSTTSLEPCGSNNNOPLSDLP :	93
MinSP1:	MTMEPN-PTSDHILDWLEGSVSFFPSFLDDFYNN-GYIHEYEFWQNDISNQYQIDANTNSS-NATNSTTNIVAASSTTTSTTSLEPN-SFNNIPFSDLP :	96
GmNSP2a:	MEID-----MDMDFSG-----YSTITNTIFSSDDDYGCNNWHWSP-----VVNWDAFTGAHDDFHLLDSIM :	57
GmNSP2b:	MDIDAI-----HNDMDFSG-----YSTITNTIFSSDDDYGCNNWHWSP-----VVNWDAFTGAHDDFHQIIDSIM :	59

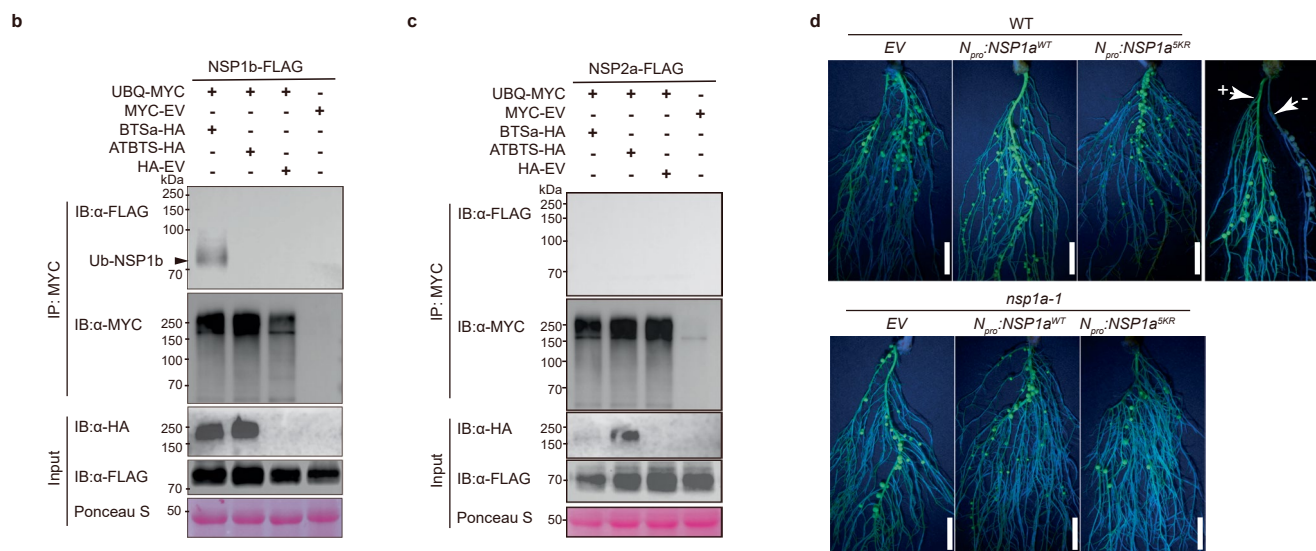
GmNSP1a:	KKRNATDESSPK-PPQNHHKKIKSKPINNEAYTGHAEGTTTT--TVRKSGGNNKKGAKANGNNCNN---KGRWAEQILNPCAATGGNLRVQHLSY :	181
GmNSP1b:	KKRSATDESSPK-PPQNHHKKIKSKPRNNEADNGDAEATTTTTATVRKSGGNNKKGAKANGNNCNN---KGRWAEQILNPCAATGGNLRVQHLSY :	181
LjNSP1:	KKRNATDESSLK-PPQN-KNKRIKTRPMN-EPENGDA-----VRK---NKKGGAKANGSNCNSGNSKEGRWAEQILNPCAATAGGNVRVQHLSY :	178
MinSP1:	KKRNAEDELSLKKQPPQNQKNRLKSRPMN-ESDNGDAALEGTV---VRKSGGNNKKGAKANGSNNNGNNKGRWAEQILNPCAATGGNLRVQHLSY :	192
GmNSP2a:	CDSAAAEEDNLS--PDDHAASNSPSASVTEEDDDADEETGPV---DSSKGLRVVHLLMAAAEALTGAPKSRDLARVILVRLKELVSHAAPPHGSNMR :	151
GmNSP2b:	CDGAAPEDNLS--PDDHVASNSPSVSITEEDDDAG-----DSSKGLRVVHLLMAAAEALSAPKSRDLARVILVRLKELVSHAAPPHGSTMR :	146

GmNSP1a:	VLHELASPTGDANHRLAAHGLKALTCHLSSSP---SSGSITFASSEPRFFQKTLKFYEVSPWFSPNNIANASTLOVLGEDTDNNSRTLHILDIGVSH :	277
GmNSP1b:	VLHELASPTGDANHRLAAHGLKALTCHLSSSPS-TSSGSITFASAEPRFFQKTLKFYEVSPWFSPNNIANASTLOVLGEDTDN-SRTLHILDIGVSH :	279
LjNSP1:	VLHELASPTGDPNHRLAAHGLRALTCHLSSSSSSPTSSGITFASTEPFRFFQKSLKFYEVSPWFSPNNIANASTLOVLAEANITRTLHILDIGVSH :	278
MinSP1:	VLHELASTTGDANHRLAAHGLRALTCHLSSSSSS-TPSGITFASTEPFRFFQKSLKFYEVSPWFSPNNIANASTLOVLAEPPN-RTLHILDIGVSH :	290
GmNSP2a:	LAAFTDALQGLLEGASGAHNKRHHYNIITSSCGPHHRDDHNNHQSNTLAAQQLQDMSPVVKFGHFTANQALLESVAH-----BRRVHIVDYDIME :	246
GmNSP2b:	LAAFTDALQGLLEGAGGAHNNNKHHHYITSCGPHHHRDDHH-HQNDTLAAQQLQDMSPVVKFGHFTANQALLESVAH-----BRRVHIVDYDIME :	240

GmNSP1a:	GMQWPTFLEALSRRPGGP-FLVRLTVVTASSSTEND---TPFCIGPFGDNFSSRLLGFAOSMNVLQINKLDNCHLHSLNAQSVDTSDEIFVUCAQF :	372
GmNSP1b:	GMQWPTFLEALSRRAGGP-FLVRLTVVTASSSTEND---TPFCIGPFGDNFSSRLLGFAOSMNVLQINKLDNCHLHSLNAQSVDASDEIFVUCAQF :	374
LjNSP1:	GMQWPTLLDALSRSSGGP-SVRLTVVT-----ENDQNMETPFSAKPPGYNYYPRLLGFAOSKINQINRIENHSLQTLNAQSISASDEILIVCAQF :	373
MinSP1:	GMQWPTFLEALSRRPGGP-FLVRLTVVNASSSTENDQNMETPFSGIGCGDTFSSGLLGFAOSMNVLQIKKLDNCHLHSLNAQSVDTSDEIFVUCAQF :	389
GmNSP2a:	GMQWASLMQALASNKTGPPGPHRITALSRTGSGRRS-----IATVQETG---RLTATAASLGQPFSSFHHCRLDEDETFKPSLKLVRGEALVFNCLM :	337
GmNSP2b:	GMQWASLIQALASNKTGPPGPHRITALSRTGSGRRS-----IATVQETG---RLTATAASLGQPFSSFHHCRLDEDETFKPSLKLVRGEALVFNCLM :	331

GmNSP1a:	RIHQLNHNPADDERSEFLKVLRLNMEPKGVILSDNNMGCCCNCCGDFTTGFSRRVEYLWRFIDSTSSAF----KGRESDERRVMEGEAAKALTNQRE-TNEG :	467
GmNSP1b:	RIHQLNHNPADDERSEFLKVLRLNMEPKGVILSDNNMGCCCNCCGDFATGFSRRVEYLWRFIDSTSSAF----KGRESDERRVMEGEAAKALTNQRE-TNEG :	469
LjNSP1:	RIHHLNHNSPDERSEFLKVLRLNMEPKGVILSENNTGCCCGCNFAAGFTTRRVEYLWRFIDSTSSAF----KGRESDERRVMEGEAAKALTNQRE-MNEE :	468
MinSP1:	RIHHLNHNNPDERSEFLKVLRLNMEPKGVILSENNTGCCCGCNFAAGFTTRRVEYLWRFIDSTSSAF----KNRDSERRVMEGEAAKALTNQRE-MNER :	484
GmNSP2a:	NIPHLSYRAPDSVASFLSGAKAKPRILVTLVEEEVG---SSAGGFVGRFMESLHHYSAVFDSLEAGFPMQGRARALVERVFFGPRIVGSLGRLYRTGEE :	434
GmNSP2b:	NIPHLSYRAPDSVASFLSGAKAKPRILVTLVEEEVA---SIVGGFVARFMDLSHHYSAVFDSLEAGFPMQGRARALVERVFLGPRIVGSLARMG--EEEE :	426

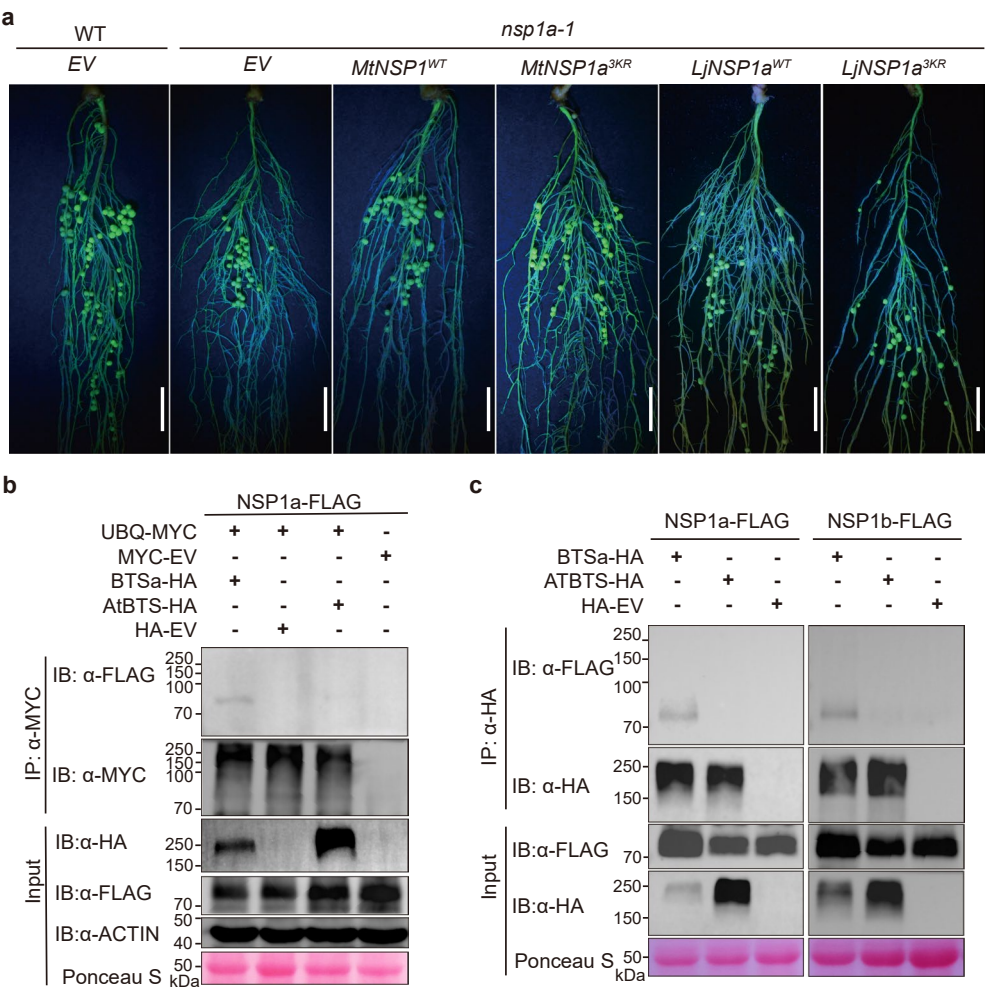
GmNSP1a:	REKWCERMKBAGFVEVFGEIDAIDGGRALLRKYDGNWEMKVEDDNTSVGLMWKQGSVSFQSLWKLDGNYKTSTFTS* :	543
GmNSP1b:	REKWCERMKBAGFVEVFGEIDAIDGGRALLRKYESNTWEMKVEDDNTSVGLMWKQGSVSFQSLWKLDGNDQSST*--- :	542
LjNSP1:	REKWCERMKBAGFVEVFGEIDAIDGGRALLRKYDGNWEMKVEENNTSVGLMWKQGPVSFQSLWKLDGNDQGGTS*--- :	542
MinSP1:	REKWCERMKBAGFVEVFGEIDAIDGGRALLRKYDGNWEMKVEENNTSVGLMWKQGPVSFQSLWKLDKQPE*----- :	554
GmNSP2a:	RGSWGWLGAGFRGVPMFSFANHCQAKLLGLFNDGYRVEELGTNK-IVLDWKSRRLLSASLWTQINSF*----- :	502
GmNSP2b:	RGSWGWLGAGFRGVPMFSFANHCQAKLLGLFNDGYRVEELGSNK-IVLDWKSRRLLSASLWTSSSHSD*----- :	495



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Monoubiquitination assays of NSP1b and NSP2a by BTSa, and complementary assays of NSP1a and NSP1a^{SKR} in *nsp1a-1* mutant. **a**, Alignment of NSP1a, NSP1b, NSP2a and NSP2b. The red stars indicate the lysine monoubiquitinated by BTSa conserved in soybean, Medicago and Lotus, and the blue stars indicate the lysine conserved in soybean and Medicago or soybean and Lotus, **b**, *In vivo* ubiquitination assay of NSP1b by BTSa or AtBTS. Proteins extracted from *N. bethamiana* leaves transiently transformed with 35S_{pro}:NSP1b-3×FLAG, 35S_{pro}:UBQ-6×MYC, UBI_{pro}:BTSa-3×HA or UBI_{pro}:AtBTSa-3×HA. Anti-MYC agarose was used for immunoprecipitation and immunoprecipitated proteins were probed with anti-FLAG and anti-MYC antibodies. The experiments were repeated for three times with similar results. **c**, *In vivo* ubiquitination assay of NSP2a by BTSa or AtBTS. Proteins extracted

from *N. bethamiana* leaves transiently transformed with 35S_{pro}:NSP2a-3×FLAG, 35S_{pro}:UBQ-6×MYC, UBI_{pro}:BTSa-3×HA or UBI_{pro}:AtBTSa-3×HA. Anti-MYC agarose was used for immunoprecipitation and immunoprecipitated proteins were probed with anti-FLAG and anti-MYC antibodies. The experiments were repeated for three times with similar results. **d**, Complementary assays of NSP1a and NSP1a^{SKR} in *nsp1a-1* mutant. Nodulation phenotype of hairy roots expressing NSP1a_{pro}:NSP1a^{WT}-4×MYC (N_{pro}:NSP1a^{WT}) or NSP1a_{pro}:NSP1a^{SKR}-4×MYC (N_{pro}:NSP1a^{SKR}) in Wm82 and *nsp1a-1* mutant. The constructs contain a 35S_{pro}:GFP expression cassette to visualize transgenic roots. + represents the transgenic roots with GFP fluorescence, - represents the non-transgenic roots without GFP fluorescence. Scar bars = 2 cm. The experiments were repeated for three times with similar results.



Extended Data Fig. 10 | Complementary assay of *NSP1*s and *NSP1^{3KR}* mutant, monoubiquitination specificity assay of *NSP1a* by *BTS*s. **a, Nodulation phenotype of transgenic soybean hairy roots in wild-type Wm82 and *nsp1a-1* heterologously expressing *UBI_{pro}:MtNSP1^{WT}-4×MYC*, *UBI_{pro}:LjNSP1^{WT}-4×MYC*, *UBI_{pro}:MtNSP1^{3KR}-4×MYC* or *UBI_{pro}:LjNSP1^{3KR}-4×MYC* constructs. Nodules were harvested at 28 DAI. Scar bars = 2 cm. The experiments were repeated for three times with similar results. **b**, *In vivo* ubiquitination assay of *NSP1a* by *AtBTS*. Proteins extracted from tobacco leaves transiently transformed with (+) or**

without (-) *35S_{pro}:NSP1a-3×FLAG*, *35S_{pro}:UBQ-6×MYC*, *UBI_{pro}:BTSa-3×HA*, *UBI_{pro}:AtBTSa-3×HA* and empty vectors. Anti-MYC agarose was used for immunoprecipitation. The experiment was repeated for three times with similar results. **c**, Co-IP assay to test the interaction of *AtBTSa-3×HA* and *NSP1a-3×FLAG* or *NSP2a-3×FLAG* in *N. benthamiana* leaves. Protein was immunoprecipitated with HA antibody-bound agarose beads, immunoprecipitated proteins were analyzed using anti-FLAG and anti-HA antibodies. The experiments were repeated for three times with similar results.

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|-------------------------------------|-------------------------------------|--|
| ☐ | ☒ | The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement |
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| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☒ | ☐ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

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Data collection bio-rad cfx manager Version 2.1 for qRT-PCR, Leica TCS SP8 was used to perform confocal microscopy. Image J was used to calculate the protein bind intensities.

Data analysis Graphpad Prism 9, Image J 1.54g, MEGA-X, TBtools

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Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

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Population characteristics N/A

Recruitment N/A

Ethics oversight N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size Sample size was determined based on previous publications on similar experiments: Chen et al., The B-type response regulator GmRR11d mediates systemic inhibition of symbiotic nodulation. Nat Commun, 2022;13(1):7661

Data exclusions No data were excluded.

Replication Each experiment was successfully repeated independently three times.

Randomization For each independent experiment, transgenic plants were randomly allocated to either the treatment or control groups.

Blinding The research materials are plants so the blinding design is not applicable to this system. Experiment results are not subjective.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☐	☒ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used anti-MYC (SIGMA, A7470) were used in CHIP assays, anti-HA (SIGMA, SAB4300603, 1:2500), anti-ACTIN (AC009, 1:3000, Abclonal), anti-GFP (ABclonal, AE012, 1:2500) and anti-MYC (SIGMA, C3956, 1:3000) were used in western-blot assays, anti-GST (ABclonal,

AE001, 1:3000) for free degradation assay. Goat anti-Rabbit IgG (HRP); NSP1 endogenous antibody (1:1000) was purified from rabbit serum after immunization with NSP1a and NSP1b-specific peptides (amino acids 4-19) .

Validation

anti-NSP1a antibody was validated by western blotting from wild type and nsp1a mutant. The other antibody we used are all from mature commercial company (SIGMA and ABclonal). The validation experiments can be found on their company websites. NSP1a

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Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

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☒	☐ National security
☒	☐ Crops and/or livestock
☒	☐ Ecosystems
☒	☐ Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes
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☒	☐ Confer resistance to therapeutically useful antibiotics or antiviral agents
☒	☐ Enhance the virulence of a pathogen or render a nonpathogen virulent
☒	☐ Increase transmissibility of a pathogen
☒	☐ Alter the host range of a pathogen
☒	☐ Enable evasion of diagnostic/detection modalities
☒	☐ Enable the weaponization of a biological agent or toxin
☒	☐ Any other potentially harmful combination of experiments and agents

Plants

Seed stocks

Soybean Glycine max (L.) Merrill c.v. Williams 82 were used in these experiments.

Novel plant genotypes

Novel plant genotypes were obtained in response to successful cas9 editing activity. The novel genotypes were produced by Agrobacterium-mediated cotyledon-node transformation.

Authentication

Novel plant genotypes were assessed by PCR-based gene amplification and Sanger sequencing.